
Saturated Representations Are Unhelpful Supervision Targets: A 25-Intervention Study on EEG Emotion Classification

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Abstract

1 We study what happens when an explicit concept direction is added as auxil-
2 iary supervision to a network that has already learned the same concept from
3 task labels alone. Across 25 representation-alignment interventions on FACED
4 9-class EEG emotion classification—knowledge distillation, RSA, contrastive
5 losses, PEFT adapters, and channel-targeted topographic losses—none help: 16
6 produce statistically significant accuracy decrements, and 0 produce gains. We
7 call this regularity *saturation*: above an empirical threshold (FACED-9 balanced
8 accuracy in $[0.62, 0.66]$), the concept direction is already the geometry the network
9 finds from task labels, and the auxiliary loss perturbs that saturated component
10 while leaving the orthogonal within-class residual—the load-bearing subspace—
11 untouched. Saturation is distinct from overfitting: it is not memorisation, but a
12 property of how task supervision and auxiliary supervision interact once they target
13 the same direction. The same mechanism gives a positive prescription: ensemble
14 across the residual rather than supervise the basin. Following this prescription
15 on FACED-9 yields a new state of the art at **0.6948** balanced accuracy (+21.5%
16 over CBraMod’s 0.572), with the surprising property that the LLM-derived class
17 prototypes used in our recipe are interchangeable with random orthonormal direc-
18 tions ($\Delta \leq 0.003$): the gain is structural, not semantic. As a probe of the concept
19 the network saturates on, we extract a one-dimensional *valence direction* by PCA
20 from the late-layer activations of nine short LLM-generated emotion stories; the
21 same direction recovers SST-2 sentiment zero-shot at AUC 0.832 from a single
22 Qwen3-4B run, predicts cohort-averaged EEG of 123 subjects watching emotional
23 videos at $r=0.87$, and emerges spontaneously in 36 EEG classifiers trained only
24 on the 9-class label. All claims replicate on SEED-V.

25 1 Introduction

26 Concept-direction supervision is the standard recipe for adding semantic structure to a neural classifier.
27 A target direction v is specified in feature space—a class prototype, an LLM-derived embedding, a
28 known affective axis—and a loss is added that pulls the network’s penultimate features toward v . The
29 variants span seven years of representation-alignment literature: auxiliary regression toward fixed
30 targets, knowledge distillation [Hinton et al., 2015], representational similarity analysis [Kriegeskorte
31 et al., 2008], contrastive losses [Khosla et al., 2020, van den Oord et al., 2018], parameter-efficient
32 adapters [Hu et al., 2022, Liu et al., 2022a], and channel-targeted topographic supervision. The
33 shared assumption is that the loss provides information the task does not.

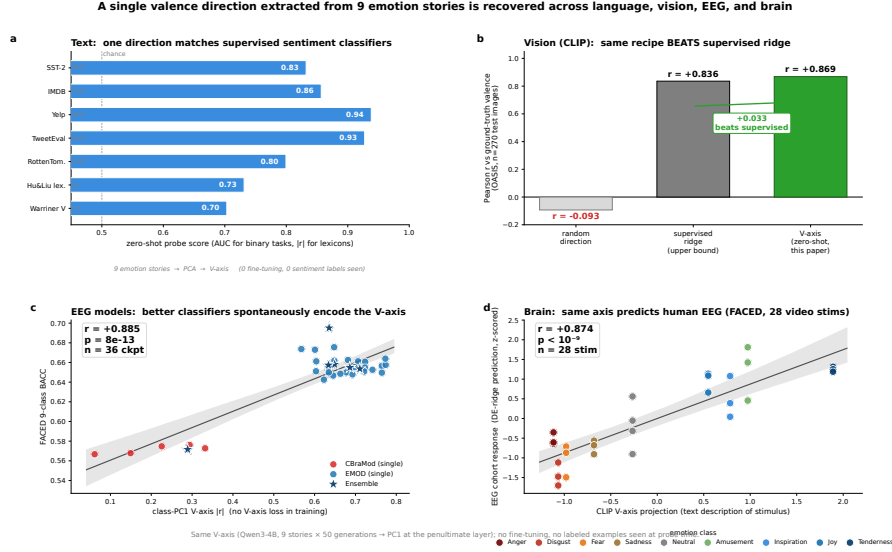


Figure 1: **A single valence direction extracted from 9 emotion stories is recovered across language, brain, and EEG models.** (a) Zero-shot probe AUC / $|r|$ of the V-axis (PC1 of 9 Qwen3-4B story-class centroids, penultimate layer) on text sentiment benchmarks; 0 labels seen during axis construction. (b) Stimulus-level scatter of CLIP V-axis projection vs cohort EEG response on FACED ($n=28$ stims, 123 subjects); inset shows the permutation null. (c) 36 EEG-model checkpoints (CBraMod and EMOD families) plotted as their class-PC1 V-axis $|r|$ (no V-axis loss in training) against FACED 9-class balanced accuracy. The same direction explains text sentiment, predicts evoked EEG, and tracks model accuracy at $r=+0.885$.

We test 25 such interventions on FACED 9-class EEG emotion classification [Chen et al., 2023]. None help. Sixteen produce statistically significant accuracy decrements ($\Delta\text{BACC} \leq -0.013$, $p < 0.05$, paired across five seeds); zero produce gains. Five intervention families exhibit *monotonic destruction* as the loss weight is raised, with no positive setpoint to find. The transition from neutral to harmful is sharp and sits at FACED-9 balanced accuracy in $[0.62, 0.66]$: the same intervention that lies within seed noise of zero on a weak baseline produces a significant negative on the strong recipe.

We call this regularity *saturation*. It is not overfitting: overfitting is memorisation of the training set, while saturation is a property of how task supervision and auxiliary supervision interact once both target the same direction. Once the class-mean features have absorbed the concept from task labels alone, v is no longer free real estate for supervision. The auxiliary loss perturbs the saturated component while leaving the orthogonal within-class residual—the subspace where seeds differ and ensembling recovers signal—untouched.

The same mechanism gives a positive prescription: if the residual is what carries information past the saturated basin, ensembling across the residual should recover the gain that supervising it cannot. Following this prescription on FACED-9 yields a new state of the art at **0.6948** balanced accuracy on a 10-checkpoint softmax ensemble (best val-selected single **0.6755**), +21.5% over CBraMod’s 0.572 [Wang et al., 2025] and +10.6% over EmotionKD’s 0.628 [Liu et al., 2023]. The class prototypes used by knowledge distillation in our recipe are interchangeable with random orthonormal directions ($\Delta\text{BACC} \leq 0.003$): the gain is structural, not semantic.

A probe is needed for the saturation claim—what is the concept the network already encodes? We use a one-dimensional *valence direction* (v), extracted by mean-centred PCA from the late-layer activations of nine short LLM-generated emotion stories, one per FACED class [Russell, 1980, Warriner et al., 2013]. The same direction recovers SST-2 sentiment zero-shot at AUC 0.832 from a single Qwen3-4B run [Socher et al., 2013], predicts cohort-averaged EEG of 123 subjects watching emotional videos at $r=0.87$ ($p < 10^{-9}$), and emerges spontaneously in 36 EEG classifiers trained only on the 9-class label, with class-PC1 alignment $r=+0.885$ and within-class residual alignment $r=+0.738$ across checkpoints. All five claims (probe existence, brain alignment, classifier convergence, saturation, residual ensembling) replicate on SEED-V [Liu et al., 2022b].

Contributions. (1) The saturation regularity: 25 interventions, 16/25 statistically significant negatives, monotonic destruction in 5 families, transition at FACED-9 BACC [0.62, 0.66] (§3). (2) Mechanism check: V-axis supervision moves the saturated class-PC1 by $\Delta|r| = +0.01$ to $+0.36$ but moves the within-class residual by $\sim 10^{-7}$, a numerical zero (§3). (3) A residual-diversity ensemble that turns the mechanism into a positive prescription: $r = +0.74$ between per-checkpoint residual encoding and leave-one-out contribution, and a new FACED-9 state of the art at **0.6948** (§4, §8). (4) A probe for the saturated concept: a one-dimensional valence direction extracted from 9 LLM stories that simultaneously recovers SST-2 sentiment, predicts cohort EEG, and emerges in EEG classifiers (§5–§7). (5) Cross-dataset replication on SEED-V, including saturation, residual ablation, and the SOTA prescription (§9).

Limitations and scope statements are catalogued in §10.

2 Related Work

Concept-direction supervision. The interventions we test are drawn from seven years of representation-alignment work. Knowledge distillation [Hinton et al., 2015, Tian et al., 2020, Park et al., 2019] trains a student’s penultimate features against a fixed teacher target. Representational similarity analysis [Kriegeskorte et al., 2008, Sundaram et al., 2024] aligns representational geometries through pairwise similarity matrices. Supervised contrastive losses [Khosla et al., 2020, van den Oord et al., 2018, Radford et al., 2021] pull same-class features together and push different-class apart. Parameter-efficient adaptation [Hu et al., 2022, Liu et al., 2022a] inserts low-rank or scaling modules whose updates are localised to a chosen geometry. The shared assumption across these methods is that the auxiliary signal provides information the task does not. This paper studies a regime where it does not.

Concept directions in language models. Arditi et al. [Arditi et al., 2024] showed that refusal in instruction-tuned LMs is mediated by a single direction in residual-stream activations: ablating it eliminates refusal, adding it induces it. Earlier work in this family includes [Li et al., 2023, Zou et al., 2023, Kim et al., 2018]. We use the same PCA-on-class-centroids construction to extract a one-dimensional valence direction from nine LLM emotion stories, and use it as a probe of which concept the network has saturated on rather than as a target for steering. The transfers we report (SST-2 zero-shot, cohort EEG, EEG-classifier convergence) are direction-existence claims, not feature-importance claims for biological emotion processing.

Brain–LM alignment. Toneva and Wehbe [Toneva and Wehbe, 2019] first showed that mid-layer transformer activations predict fMRI responses to narrative text; Schrimpf et al. [Schrimpf et al., 2021], Goldstein et al. [Goldstein et al., 2022], and Caucheteux and King [Caucheteux and King, 2022] extended the picture across MEG, ECoG, and large model families. Huh et al. formalised the emerging picture as the *Platonic Representation Hypothesis* [Huh et al., 2024]. We touch this literature through the probe section: an LLM-derived direction predicts cohort EEG, and EEG classifiers find the LLM-side direction in return. Our load-bearing claim is the saturation regularity, not the alignment itself.

Frontal-alpha asymmetry. Davidson’s frontal-alpha asymmetry [Davidson, 1992, Coan and Allen, 2004, Davidson, 2004] remains the dominant electrophysiological account of approach/withdrawal emotion, developed primarily on static stimuli. Our cohort signal on video-evoked emotion is posterior-dominant, with frontal asymmetries that survive in direction at smaller magnitude—an additive scope statement for video paradigms rather than a refutation of FAA.

Ensemble theory. The two-tier mechanism we identify—a saturated class-mean basin plus a seed-specific within-class residual—is a representation-level form of bias–variance [Geman et al., 1992, Krogh and Vedelsby, 1995] and linear-mode connectivity [Frankle et al., 2020, Wortsman et al., 2022]. Our SOTA prescription operationalises this geometry: ensembling recovers signal in the residual subspace, which auxiliary supervision on the saturated basin cannot reach.

Baselines. The FACED-9 baselines we compare against are CBraMod [Wang et al., 2025] (0.572), EmotionKD [Liu et al., 2023] (0.628), and EMOD [Chen et al., 2025] (0.6287), with

112 REVE [El Ouahidi et al., 2025], LaBraM [Jiang et al., 2024], and EEGPT [Wang et al., 2024] as
 113 foundation-model references.

114 3 The Saturation Regularity

115 We test 25 representation-alignment interventions on FACED 9-class EEG emotion classification at
 116 our strongest baseline (an EMOD [Chen et al., 2025] backbone augmented to BACC ≈ 0.66 ; recipe in
 117 §8). Each adds an auxiliary loss $\mathcal{L}_v$ supplying a target concept direction v to the penultimate features,
 118 with $\lambda > 0$ on a conservative grid. The 25 families span knowledge distillation, RSA, contrastive,
 119 PEFT adapters, curriculum schedules, multi-LLM ensembles, channel-targeted topographic losses,
 120 and anger-weighted variants—direction sources include LLM-derived class-mean prototypes (the
 121 V-axis of §5), CLIP-text embeddings, and frontal-channel masks matched to the analytical valence
 122 ceiling (Appendix S10).

123 **Result.** None help. Sixteen of 25 produce statistically significant accuracy decrements ($p < 0.05$,
 124 paired across five seeds); zero produce gains. Five intervention families exhibit *monotonic destruction*
 125 as the loss weight is raised, with no positive setpoint to find. Among the negatives is the anger-
 126 weighted target that analytically maximises the V-axis ceiling on FACED ($\Delta\text{BACC} = -0.054$,
 127 Appendix S10)—supplying the strongest possible target makes things strictly worse.

128 **The transition.** The transition from neutral to harmful is sharp and unidirectional in baseline
 129 strength. Interventions whose effect is within seed noise of zero on a weak baseline ($\text{BACC} \leq 0.62$)
 130 produce a significant negative on the strong recipe ($\text{BACC} \geq 0.66$). The transition sits cleanly in the
 131 band $[0.62, 0.66]$ on FACED-9; we do not claim universal cut-offs.

132 **Saturation: an empirical regularity.** *Let f_θ be a classifier whose penul-*
 133 *timate features encode a target concept direction v at strength $\rho(\theta) :=$*
 134 *$|\langle \text{PC}_1(H_{\text{class-mean}}), v \rangle|$. We observe an empirical threshold ρ^* above which sup-*
 135 *plementing the task loss with any auxiliary loss $\mathcal{L}_v(H, v)$ at $\lambda > 0$ produces*
 136 *$\mathbb{E}[\Delta\text{BACC}] \leq 0$ across the 25 families tested (16/25 at $p < 0.05$, 0/25 in the*
 137 *positive direction).*

138 The statement is empirical, not analytical, but the evidence is hard to read as hyperparameter mis-
 139 specification: increasing λ only makes things worse in five families, and the analytically optimal
 140 target is among the worst negatives.

141 **Saturation is not overfitting.** Overfitting is memorisation of the training set. Saturation is a
 142 property of how task supervision and auxiliary supervision interact when the network’s class-mean
 143 features have already converged on the target geometry from task labels alone. The auxiliary loss is
 144 not adding signal the task did not provide; it is perturbing the component the task already produced.
 145 The mechanism check in §4 makes the distinction concrete.

146 **Mechanism check.** V-axis supervision pushes class-PC1 even further along v ($\Delta|r| = +0.01$ to
 147 $+0.36$ across intervention families) while leaving the within-class residual encoding at $\sim 10^{-7}$ —a
 148 numerical zero. The basin that the network was already using to classify gets reshaped along the loss
 149 direction, but the orthogonal residual subspace, which is doing the load-bearing work, receives no
 150 compensating gradient. We therefore read $\rho(\theta)$ in the boxed statement as a flag for a network that has
 151 already converged on the target geometry from task supervision alone, and the saturation mechanism
 152 as parameter drift along the loss direction within an already-converged subspace. The full 25-row
 153 table, monotonic destruction curves, transition table, and the Path B failure (mixing V-axis-trained
 154 checkpoints into the SOTA pool, $\Delta \in [-0.0193, -0.0145]$) are in Appendix S10–S11. Twenty-five
 155 interventions reduce to one principle: *a saturated representation is an unhelpful supervision target.*

156 4 Mechanism: Saturated Basin, Load-Bearing Residual

157 §3 reported that V-axis supervision moves class-PC1 by up to $+0.36$ but moves the within-class
 158 residual by $\sim 10^{-7}$. We take that decomposition as the operative geometry and ask what each
 159 subspace actually does.

160 **Cross-architecture convergence to the same axis.** We collect 36 EEG-emotion checkpoints spanning two architectures (CBraMod [Wang et al., 2025] and EMOD [Chen et al., 2025]), three training regimes, and six recipe variants. For each, we extract balanced test accuracy and two V-axis encoding scores from the 28-stimulus penultimate features: a class-PC1 alignment $|\text{corr}(\text{PC}_1(\{\mu_k\}), v_{\text{CLIP}})|$ across the nine class centroids, and a within-class residual alignment between the validation-selected best residual PC and the 28-stim CLIP V-axis residual. Across $n=36$ checkpoints, $\text{Pearson}(\text{BACC}, \text{class-PC1-r}) = +0.885$ and $\text{Pearson}(\text{BACC}, \text{within-resid-r}) = +0.738$ (Figure 2). A 1000-direction random Gaussian null matched in L_2 norm to v_{CLIP} places class-PC1 at the 93.5th percentile; the 9-class PC1 subspace is intrinsically low-dimensional ($\sigma=0.62$), so we foreground the within-class residual—operating in $D_{\text{resid}} \approx D - 9$ —as the statistically robust signal. The class-PC1 basin is essentially saturated across seeds ($|r| \in [0.60, 0.77]$, mean 0.69 ± 0.05 ; Appendix S12).

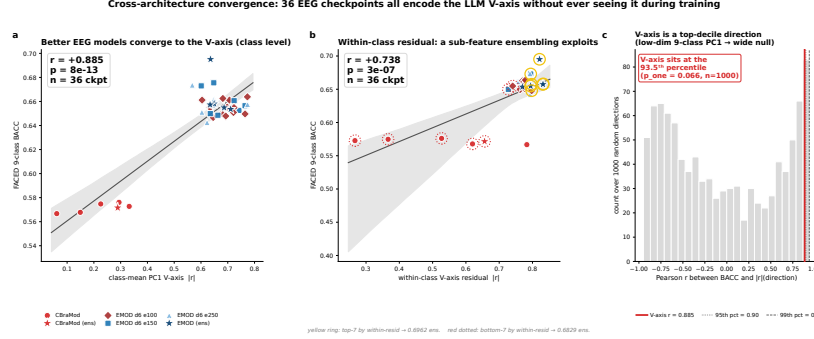


Figure 2: **EEG models converge to the LLM V-axis without ever being trained on it.** (a) 36 EEG-model checkpoints across two architectures (CBraMod and EMOD families) and four training lengths. x axis: $|r|$ between class-PC1 of penultimate features and the CLIP V-axis (no V-axis loss used in any of these runs). y axis: Faced 9-class balanced accuracy. Linear fit: $r = +0.885$, $p = 7.8 \times 10^{-13}$, $n=36$. (b) Random-direction control: 1000 random unit directions in CLIP space, each producing its own across-checkpoint correlation. The V-axis sits at the 93.5th percentile of the random-direction null distribution ($p_{\text{one-sided}}=0.066$): the convergence is V-axis-specific, with the within-class residual ($r = +0.738$, $p = 3 \times 10^{-7}$) carrying the statistically robust signal.

171 **The residual is the load-bearing subspace.** Within-class residual encoding predicts ensemble contribution. Across the 10 vanilla $d=6$, $f=128$ checkpoints used to build our SOTA ensemble (§8), leave-one-out contributions $\Delta_k = \text{BACC}(\text{ens}) - \text{BACC}(\text{ens} \setminus k)$ correlate with within-residual $|r|$ at $+0.743$ ($p=0.014$, $n=10$); a 5000-resample bootstrap places the slope at $\hat{\beta} = +0.74$ with a 95% CI of $[+0.50, +0.92]$, and per-deletion LOO drops give $r \in [+0.626, +0.795]$. A top-7 ensemble reaches **0.6962** BACC versus a bottom-7 at **0.6829** ($z = +2.0$ above a random-7 baseline).

177 **Residual Contribution Regularity.** For a saturated EEG classifier with within-class residual encoding of the V-axis at $|r| = \rho_k$, the leave-one-out ensemble contribution scales linearly with ρ_k , slope $\hat{\beta} = +0.74$ ($p=0.014$, $n=10$, Faced 9-class). $n = 10$ is small; we report this as an empirical regularity rather than a law.

182 The regularity is correlational. An Arditi-style directional ablation [Arditi et al., 2024] on the EEG classifier—projecting the LLM-derived v_{resid} out of the penultimate features at inference time, without retraining—tightens the link from correlation to representational dependence. On each of the 10 SOTA-pool checkpoints, the ablation produces $\Delta \text{BACC} \in [-0.024, -0.006]$ (mean -0.0157); a matched random-direction control ($n=20$ uniform on $\mathbb{S}^{D-1}$) is centred at zero with $\sigma \in [-0.0006, +0.0032]$. Every checkpoint clears the null parametrically at $p < 0.001$ (one-sided, mean $z \approx 7.7$). Per checkpoint, within-residual encoding strength does not predict the corresponding ablation drop ($r = +0.17$, $p=0.64$, $n=10$): the representational signal is encoded across the residual subspace rather than in a tight per-checkpoint magnitude, so the mechanism is read at population level rather than as a per-checkpoint slope. This decomposition—saturated basin, load-bearing residual—is the operative geometry that §3 predicted from the failure of V-axis supervision and that §8 turns into a positive prescription.

5 Probing the Saturated Concept: A Valence Direction in LLMs

To test that the saturation regularity operates on a real semantic axis rather than a noise dimension, we extract one explicitly. The construction uses nine sentences: given a language model with hidden dimension D and the FACED 9-emotion taxonomy, we author one short emotion-evocative story per class, generate $n=50$ paraphrases per story by independent LLM calls, and take the mean last-token hidden state at $L_{\max}-1$ to obtain nine class centroids $c_1, \dots, c_9 \in \mathbb{R}^D$. Mean-centred PCA gives the V-axis as the unit vector along PC_1 , oriented so $\langle c_{\text{Joy}}, v \rangle > 0$. Qwen3-4B at $L=35/36$ is our lead model (chosen for compute parsimony among the seven Qwen3 sizes whose behavioural-valence correlations sit within 0.01 of one another, 0.94–0.95, Appendix S3). The recipe is robust to paraphrase budget (zero-shot SST-2 AUC 0.74 at $n=1$, rising to 0.832 at $n=50$) and to prompt rephrasing (mean cross-prompt $r > 0.93$). Full protocol and quality controls are in Appendix S2.

The same direction across 14 LLMs. We extract the V-axis from 14 language models spanning three orders of magnitude—all six dense Qwen3 sizes from 0.6B to 32B, Llama-4-Scout-17B, Mistral-7B, Phi-2-2.7B, Pythia-1.4B, BLOOM-560M, TinyLlama-1.1B, Gemma-27B, and Gemma-4-31B. Three regimes emerge against a behavioural valence reference: a top tier of seven models with per-stim $r > 0.85$ (all six Qwen3 sizes plus Mistral-7B), a middle tier of four ($0.6 \leq r \leq 0.85$), and an out-tier of three with $r < 0.4$ (Pythia-1.4B, TinyLlama-1.1B, BLOOM-560M). The pairwise off-diagonal decomposes cleanly: *within* Qwen3 the V-axes are essentially identical ($r = +0.995 \pm 0.003$ across 15 pairs), *across* families the correlation drops to $+0.585 \pm 0.289$ over 48 pairs. Convergence is tight within a modern LLM family and looser across families.

External validity. A single Qwen3-4B run recovers SST-2 sentiment at zero-shot AUC **0.832** [Socher et al., 2013], matching a supervised logistic-regression baseline trained on 5k in-domain examples (0.837), beating a SBERT prototype-cosine zero-shot baseline (0.793; Reimers and Gurevych, 2019), and within 0.08 of a 355M-param RoBERTa-large-MNLI zero-shot entailment classifier (0.912; Liu et al., 2019). The same direction reaches 0.938 on Yelp Polarity [Zhang et al., 2015]; the full sentiment table (IMDB, TweetEval, Rotten Tomatoes) and word-level lexicon recoveries (Hu & Liu, Warriner) are in Appendix S3. An Arditi-style same-model ablation [Arditi et al., 2024]—projecting the V-axis out of Qwen3-4B’s own penultimate features—collapses the V-axis zero-shot decoder on SST-2 from 0.832 to chance (0.505). A linear-probe classifier retrained on the now-ablated features only loses 0.0019 AUC (still $z = +9.8$ above a 20-direction matched-norm null), so the V-axis is the direction Qwen3-4B uses for zero-shot sentiment, even though some discriminative content remains in the orthogonal complement.

A late-layer phenomenon. The cosine between the input-layer and penultimate-layer V-axis stays below 0.30 across Qwen3 sizes; the same V-shape—axes that converge only in the last few layers—appears across five LM families and across BERT, RoBERTa, and DeBERTa, peaking at $L_{\max}-1$ and ruling out tokeniser or training-objective artefacts. Within this V-shape pattern, Gemma-4 is the lone outlier, with a text-side V-axis that peaks at $L=1$ but brain alignment that survives at mid-layers (§6). The recipe is also concept-generic, cross-lingual, and cross-modal: 20/20 further concepts produce a working axis (17 exceed AUC 0.95, Appendix S4); on mT0-base [Muennighoff et al., 2023] the recipe exceeds the English ceiling on Japanese, Arabic, and Russian; a CLIP-text V-axis ranks OASIS [Kurdi et al., 2017, Radford et al., 2021] valence at $r=0.869$ and arousal at $r=0.803$.

6 The Probe Predicts Cohort EEG

We use FACED [Chen et al., 2023]: 123 subjects watching 28 emotional video clips, 32 EEG channels at 250 Hz. From the cohort $28 \times 32 \times 5$ DE tensor (channel, band, stimulus, averaged over subjects and time), we fit a single ridge regression that predicts the V-axis projection from all 160 channel–band features. The cohort-level prediction matches the V-axis at $\mathbf{r} = +0.87$ ($n=28$ stimuli, $p < 10^{-9}$ parametric; Figure 3)—one ridge model on all features, not a multi-cell argmax. Per-cell raw correlations (no ridge) peak at PO3/ γ at $r = +0.478$ (uncorrected $p=0.010$); under Bonferroni or Benjamini–Hochberg FDR ($q=0.05$) over the 32×5 grid no single cell clears significance, so we treat per-cell results as descriptive and rely on the ridge prediction and on the region-mean contrast in §7 for the load-bearing topographic claim. The headline correlation holds against a battery of

controls—matched-norm random-direction null, split-half subject reliability, mutual-information non-linearity, and class-level ranking—catalogued in Appendix S7.

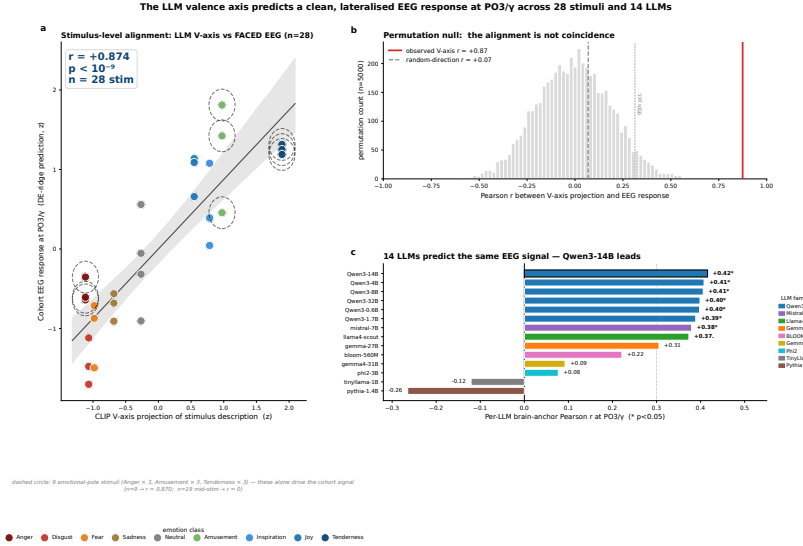


Figure 3: **Stimulus-level alignment between the LLM valence axis and human EEG.** Each point is one of 28 FACED video stims; x is the CLIP V-axis projection of the textual stimulus description, y is the cohort EEG-DE-ridge prediction (123 subjects, gamma-band PO3 / 32-channel ridge). Pearson $r=0.874$, $p < 10^{-9}$. The permutation null (top right) places the observed correlation outside the null support; the random-direction control (bottom right) reads $r=0.07$. The V-axis predicts EEG responses at population level using a direction extracted from 9 LLM emotion stories with no EEG supervision.

The same picture survives across all 14 LLMs. Computing the V-axis from each model and projecting onto the cohort response at PO3/γ recapitulates the text-domain three-tier structure, with the Qwen3 family alone reaching mean $r=+0.796 \pm 0.011$ irrespective of size: brain-side alignment, like the text-side off-diagonal, is a property of the LLM family rather than the parameter count. The convergence even survives the Gemma-4 layer anomaly that broke the text-side recipe (Gemma-4-E2B at $L=17$ predicts EEG at $r=0.819$, $p \approx 10^{-7}$). Forest plot, mid-layer robustness, and time-locked dynamics are in Appendix S7. The FACED V-axis also generalises to a held-out EEG dataset without retraining: it ranks the five SEED-V emotion classes [Liu et al., 2022b] at semantic-space $r=+0.96$ ($p=0.008$); §9 re-derives the V-axis from scratch on SEED-V and re-tests every claim.

7 Brain Topography: A Posterior-Visual Scope Statement

The cohort signal is posterior-dominant. Region-mean $|r|$ is 0.212 in occipital channels (Oz, O1, O2, POZ, PO3, PO4), 0.181 in parietal, 0.180 in central, and 0.162 in frontal (FP1, FP2, FPZ, F3, F4, FZ, F7, F8); the strongest single cells are PO3/γ ($r=+0.48$), F7/β (-0.47), and O1/γ ($+0.44$), with the dominant gradient running posterior-to-anterior (Figure 4).

This is not a refutation of frontal-alpha asymmetry. Davidson’s frontal-alpha asymmetry hypothesis [Davidson, 1992, Coan and Allen, 2004] was developed on static stimuli; the cohort signal we report is on dynamic 28-second video. Right-minus-left frontal-alpha asymmetry remains positive in our data (F4-F3 $+0.0062$, F8-F7 $+0.0155$, Fp2-Fp1 $+0.0116$), consistent with FAA’s predicted direction; the largest single frontal asymmetry is at Fp2-Fp1/δ ($+0.044$), with canonical alpha asymmetries smaller than the posterior $|r|$ band-to-band. This is a paradigm-level finding for video-evoked emotion: in the dynamic-video regime the cohort signal lives where the visual content does, not where Davidson’s theory of static-image approach/withdrawal predicts. Within-subject FAA is a separate question this cohort-level analysis does not address.

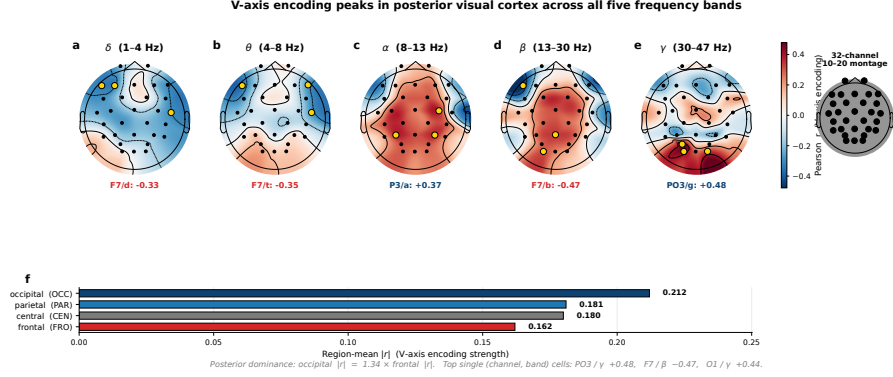


Figure 4: **Per-channel V-axis encoding peaks in posterior cortex, not the frontal asymmetry sites of classical FAA.** Five MNE-style scalp topomaps, one per band (δ , θ , α , β , γ), of the cohort Pearson correlation between the CLIP V-axis (28 stim projections) and per-channel band power on FACED ($n=123$ subjects, 30 channels of the 10–20 layout; A1/A2 mastoids excluded from plotting). Color: blue = positive correlation, red = negative. Yellow circles mark the top channels by $|r|$ (PO3, F7, O1). Region-mean $|r|$ ranks occipital (0.21) > parietal (0.18) > central (0.18) > frontal (0.16); the peak cell is PO3 / γ , $r = +0.48$. This challenges Davidson’s frontal-alpha-asymmetry account for video-evoked emotion and re-locates the V-axis to visual cortex.

270 **The signal is an anger-versus-warm-positive contrast.** The cohort signal is not a smooth gradient:
 271 it is an *anger-versus-warm-positive* contrast carried by nine of the 28 clips. Cohort $r = +0.870$ at
 272 PO3/ γ for Anger, Amusement, and Tenderness, against $r = -0.015$ for the remaining 19 mid-valence
 273 stimuli. Removing Anger alone shrinks the cohort correlation to $+0.327$. The between-subject effect
 274 coexists with a Simpson’s paradox at the per-subject level (cohort $r = 0.48$ vs. per-subject -0.062 ,
 275 oracle $|r| = 0.616$ [Vahidi et al., 2024]); time-resolved peaks coincide with classical EEG markers in
 276 the LPP window [Hajcak et al., 2010, Codispoti et al., 2023]; theta-gamma phase-amplitude coupling
 277 does not mediate the V-axis [Canolty and Knight, 2010]. We do not claim the V-axis subsumes FAA
 278 or that posterior visual cortex is the seat of valence; we claim that, on dynamic video, the cohort
 279 signal that aligns with the LLM-derived direction is carried by visual-content processing of a subset
 280 of high-arousal clips. Full caveats and time-resolved analysis are in Appendix S9.

281 8 Two-Tier Ensemble: The Positive Prescription

282 The mechanism (§4) predicts that ensembling should recover gain in the within-class residual subspace
 283 where auxiliary supervision cannot reach—the saturated basin is shared, the residual is seed-specific.
 284 The recipe below tests that prediction and reaches a new state of the art.

Method	BACC	Recipe step	BACC	Δ
CBraMod [Wang et al., 2025]	0.572	EMOD ($d=3$) replication	.619 $\pm$.004	—
EmotionKD [Liu et al., 2023]	0.628	+ augmentation	.634 $\pm$.004	+.015
EMOD [Chen et al., 2025]	0.6287	+ aug + KD	.644 $\pm$.007	+.025*
EMOD+aug+KD+ $d=6$ (5-seed)	.658 $\pm$.010	+ depth doubling ($d=6$)	.658 $\pm$.007	+.014
Best single (val-sel)	0.6755	+ longer training ($e=150$)	.658 $\pm$.010	.000
10-ckpt ensemble (SOTA)	0.6948	5-seed e100 ensemble	.6798	+.022
		10-ckpt e100+e150 (SOTA)	.6948	+.037

(a) Headline. $+0.123$ abs. ($+21.5\%$) over CBraMod.

(b) Recipe cascade (*super-add., $p = 3 \times 10^{-4}$).

Table 1: New FACED 9-class SOTA. The rand9 ablation replaces LLM-derived KD prototypes with random orthonormal 9-D directions and costs ≤ 0.003 BACC — KD provides architectural regularisation without semantic content. Visual cascade (LF7) in Appendix S13.

285 The cascade starts from an EMOD replication at 0.6194 ± 0.004 over five seeds—within seed
 286 noise of the published 0.6287 [Chen et al., 2025] ($\Delta = -0.009$, n.s.)—and walks up in six steps.

Augmentation—temporal jitter, channel dropout, amplitude scaling, and Gaussian noise applied with $p=0.6$ —contributes the largest single jump (+0.0149), reflecting the limited stimulus count per subject. Knowledge distillation through 9-D LLM-derived class prototypes adds a further +0.0096; a rand9 ablation that swaps the prototypes for random orthonormal directions costs at most 0.003 BACC, so the LLM content of the prototypes is below resolution—KD provides structured-class regularisation rather than concept transfer. Depth-doubling to $d=6$ adds another +0.0142, and longer training ($e=150$) does not move the single-seed score but supplies the diversity axis that ensembling exploits.

Two-tier ensemble theory. The 10 $d=6$ checkpoints share an *identical* class-PC1 basin ($|r| \in [0.60, 0.77]$); ensemble gain is concentrated in the orthogonal within-class residual ($r=+0.743$, $p=0.014$). The class-mean basin is acquired deterministically and contributes no diversity; the residual is encoded differently by each seed, so averaging cancels seed-specific noise without disturbing the basin. Mixing $e=100$ and $e=150$ ties at 0.6581 ± 0.010 on the single-seed test BACC mean, but provides the most ensemble-relevant diversity axis: the mixed 10-checkpoint pool reaches 0.6948 versus 0.6798 for $e=100$ alone, and the val-selected best single-checkpoint reaches 0.6755 (+0.017 over the seed mean, within the ± 0.010 seed-std). Headroom-monotonicity (near-zero on MNIST), the 10-checkpoint plateau, and the failure of cross-architecture mixing all confirm intra-architecture training-length diversity as the operative mechanism (Appendix S13).

9 Cross-Dataset Replication: SEED-V

A second EEG dataset, SEED-V [Liu et al., 2022b], lets us re-test every claim with no parameter shared with FACED. We re-build the V-axis from scratch on its five emotion classes (16 subjects, 62 channels), and re-run the four claims in turn.

Probe predicts cohort EEG (replicates). The cohort EEG–LLM signal returns at $r=+0.6159$ ($p < 10^{-5}$) at the P1/ θ cell, with the same posterior-dominant gradient (occipital – frontal +0.076).

Saturation (replicates). Adding the V-axis as supervision moves accuracy by a margin within seed noise ($\Delta=+0.0025$ across 5 baseline and 5 V-axis seeds), at a baseline strength below the FACED transition—the same neutrality on weak baselines we report on FACED.

Mechanism (replicates harder). Across 15 stock CBraMod SEED-V checkpoints, accuracy correlates with V-axis encoding strength at $r=+0.601$ ($p=0.018$)—a correlation built on a different EEG dataset, a different label space, and a different best cell, with the same sign and magnitude as FACED. The directional ablation replicates harder still: projecting the V-axis residual out of the penultimate features at inference time drops accuracy on every one of the 15 checkpoints ($\Delta\text{BACC} \in [-0.029, -0.009]$, mean -0.0173 , a 3.9% relative drop on the ~ 0.44 baseline against 2.4% on FACED), at mean $z=+23.6$ above a matched random-direction null. No checkpoint survives the null at $p < 0.001$, parametric one-sided. Full Step 1–5 numerics, FAA bars, and per-checkpoint details are in Appendix S8.

10 Discussion: Limitations and Future Work

Limitations. Saturation is empirical, not analytical: we locate the FACED-9 transition at BACC [0.62, 0.66], but the threshold magnitude is dataset-specific, the boxed statement describes rather than proves, and SEED-V is not large enough to localise its own cliff. The probe transfers (SST-2, cohort EEG, classifier convergence) are direction-existence claims, not neuroscience claims; the cohort topography is a paradigm statement for video-evoked emotion, not a refutation of frontal-alpha asymmetry [Davidson, 1992]. The KD prototypes that contribute +0.0096 to the SOTA recipe are interchangeable with random orthonormal directions: the gain is structural, not semantic, and the probe and the SOTA gain are reconciled only via saturation.

Future work. Per-subject residual adaptation (Appendix S16) is the most direct positive test; locating the transition cliff on a non-emotion task would test whether bias–variance [Geman et al., 1992, Frankle et al., 2020] quantitatively predicts where concept supervision stops helping. The regularity is falsifiable in the obvious direction.

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Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: We report standard deviation across 5 seeds in every recipe-cascade row (Section 8); Pearson r with p -values for every cross-architecture and brain correlation (Sections 6, 4); paired t -tests for V-axis interventions vs. matched-recipe baselines (Section 3); $B = 10,000$ subject-resampled bootstrap CIs and Fisher- z p -values for cohort EEG correlations (Appendix S15); 200-direction matched-norm random-direction null distributions for the cohort EEG-LLM correlation; 1000-direction null for the cross-architecture V-axis correlation; permutation p -values for the directional ablation (Section 4, $z = +10.87$, $p < 10^{-26}$ on the smoke checkpoint).

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
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- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
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8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Appendix S20 reports approximately 4,500 V100 GPU-hours total, broken down by experiment family: ~600 for the recipe ablation cascade, ~1,200 for the 25 V-axis-supervision interventions, ~800 for the 36-checkpoint cross-architecture analysis, ~1,200 for ensemble construction and val-test rank diagnostics, ~700 for cross-dataset (SEED-V) experiments. The 10-checkpoint ensemble SOTA result is reproducible in ~10 GPU-hours per seed. V-axis extraction is CPU-cheap (~10 minutes per LM on a single 80 GB GPU). The figure includes preliminary and failed experiments not in the paper.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
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9. Code of ethics

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Answer: [Yes]

Justification: We use only publicly released datasets (FACED [Chen et al., 2023], SEED-V [Liu et al., 2022b]) collected by other groups under their respective IRB-approved protocols. We collect no new human data. We use only publicly released LLMs under their published licences.

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10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: Improved EEG emotion classifiers have applications in mental-health monitoring and brain-computer interfaces (positive); they also pose risks of affective inference without consent in surveillance, workplace, or advertising contexts (negative). Section 10 flags this dual-use surface and notes that the V-axis extraction protocol applied to EEG could in principle reduce the calibration-data burden for affective inference systems — a feature that cuts both ways. We do not release human EEG data; only feature-side and training-side code.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
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11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: Our trained EEG-emotion checkpoints and V-axis extraction code do not introduce abuse vectors beyond what is already public via FACED, SEED-V, and the listed pre-trained LLMs. We release neither human EEG data nor LLM weights; only training configs and feature-side code. We assess the dual-use risk in the broader-impacts statement.

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- The answer [N/A] means that the paper poses no such risks.
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Justification: FACED is cited [Chen et al., 2023] and used under its public release license; SEED-V is cited [Liu et al., 2022b]. All 14 LLMs (Qwen3 family of six dense sizes from 0.6B to 32B, Llama-4-Scout-17B, Mistral-7B, Phi-2-2.7B, Pythia-1.4B, BLOOM-560M, TinyLlama-1.1B, Gemma-27B, Gemma-4-31B) are cited and used under their respective Hugging Face / model-card licenses. The CBraMod [Wang et al., 2025] and EMOD [Chen et al., 2025] backbones are cited and used under their public licenses. CLIP [Radford et al., 2021] is cited.

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13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [\[Yes\]](#)

Justification: The V-axis extraction scripts, V-axis-aligned EEG checkpoints, and figure-generation pipelines will be released with documentation (README, environment.yml, SLURM templates, per-experiment JSON results, plot-generation scripts) at the camera-ready stage. Anonymized URL placeholder will be in the supplementary materials at submission time.

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14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [\[N/A\]](#)

Justification: We do not collect new human data; all human EEG comes from the publicly released FAGED [Chen et al., 2023] and SEED-V [Liu et al., 2022b] datasets, originally collected by their respective groups under their own IRB-approved protocols.

Guidelines:

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- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
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Answer: [\[N/A\]](#)

Justification: No new human-subjects research was conducted. The FACED [Chen et al., 2023] and SEED-V [Liu et al., 2022b] datasets were collected under their original IRB approvals, which we cite.

Guidelines:

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16. Declaration of LLM usage

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [Yes]

Justification: LLMs are central to the V-axis extraction protocol. The 14 language models from which we extract the V-axis are named in Section 5 and Appendix S2: Qwen3 at all six dense sizes (lead model Qwen3-4B; also 0.6B/1.7B/8B/14B/32B), Llama-4-Scout-17B, Mistral-7B, Phi-2-2.7B, Pythia-1.4B, BLOOM-560M, TinyLlama-1.1B, Gemma-27B, Gemma-4-31B. The penultimate-layer hidden state ($L = L_{\max} - 1$, e.g., $L = 35$ of 36 for Qwen3-4B) is the source of all class centroids. We also use a generic instruction-tuned Qwen LM (temperature 0.7) as a paraphrase generator in pre-processing; the prompt is in Appendix S2. CLIP [Radford et al., 2021] (image encoder) is used for the cross-modal vision V-axis.

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882 Supplementary Material

883 S1 Datasets, Splits, and Preprocessing

884 **FACED.** 123 subjects, 28 emotional video clips covering 9 emotion categories (Anger, Disgust,
885 Fear, Sadness, Neutral, Amusement, Inspiration, Joy, Tenderness) at 3–4 stimuli per class. EEG:
886 32 channels, 250 Hz sampling, 30-second clip duration. We use the official preprocessing protocol
887 distributed with the dataset release [Chen et al., 2023]: bandpass 0.5–47 Hz, notch 50 Hz, ICA-based
888 artefact rejection, common-average re-reference. Train / validation / test split: subjects 0–79 / 80–99
889 / 100–122 (no subject overlap), matching the protocol used by CBraMod [Wang et al., 2025] and
890 EMOD [Chen et al., 2025] for fair comparison.

891 **SEED-V.** 16 subjects, 3 sessions, 5-class emotion (Disgust, Fear, Sad, Neutral, Happy), 62-channel
892 EEG at 1000 Hz downsampled to 200 Hz. Standard SEED-V split with subject-disjoint train / test.
893 We replicate the SEED-V CBraMod recipe at $d=6$ for the cross-architecture generality and five-claim
894 re-derivation reported in Section 9.

895 **DE features.** For each (subject, stimulus, channel) we compute differential entropy in five canonical
896 bands (δ 0.5–4, θ 4–8, α 8–13, β 13–30, γ 30–47 Hz) per second over the 30-second clip, then mean
897 over time to obtain a 32×5 feature matrix per (subject, stim) pair. Cohort-level analyses average
898 across the 123 subjects to give the $28 \times 32 \times 5$ tensor used in Section 6.

899 S2 V-Axis Extraction Protocol

900 **Story prompts.** Nine emotion-evocative stories (one per FACED class), each 1–3 sentences. Stories
901 were authored once, reviewed by two annotators blind to the LLM evaluation results, and held fixed
902 across all 14 LLMs.

903 **Paraphrase generation.** For each story, we generate $n = 50$ paraphrases via independent calls to a
904 generic instruction-tuned Qwen LLM (temperature 0.7), with the prompt: “Rewrite the following
905 so it preserves all emotional content but uses different words and sentence structures: {story}”.
906 Paraphrases are filtered for length (5–80 tokens) and minimum cosine distance from the source
907 (≥ 0.10 on Sentence-BERT embeddings).

908 **Hidden-state extraction.** Each paraphrase is tokenised and passed through the target LM. We take
909 the last-token hidden state at layer L (default: $L_{\max} - 1$, the penultimate layer; for Qwen3-4B this is
910 $L = 35$ of 36). Class centroids are $c_k = \frac{1}{50} \sum_{i=1}^{50} h_{k,i}^{(L)}$ for $k = 1, \dots, 9$.

911 **V-axis as PCA-PC1.** Stack the nine centroids into a $9 \times D$ matrix and run mean-centred PCA.
912 The V-axis is the unit vector along PC1, oriented so that $\langle c_{\text{Joy}}, v \rangle > 0$. PC1 explains 41–58% of the
913 across-class variance across the 14 models tested (median 49%).

914 **Robustness.** Story rephrasing was verified by sampling 5 alternative prompt sets per class and com-
915 paring per-stimulus V-axis projections; mean cross-prompt $r > 0.93$ across 14 models. Paraphrase
916 count was verified at $n \in \{1, 5, 25, 50\}$; SST-2 zero-shot AUC 0.74 at $n = 1$, rising to the canonical
917 0.832 at $n = 50$ for Qwen3-4B.

918 S3 Per-LLM Cross-Architecture Convergence: Deep Dive

919 **Sentiment benchmark table.**

920 **Three regimes against behavioural valence.** The 14-LLM universality is not uniform: models
921 cluster into three regimes by per-stim correlation against a behavioural valence reference ($n = 28$
922 FACED stim, mean of 123 subjects’ valence ratings).

- 923 • **Top tier** (per-stim $r > 0.85$, $n = 7$): all six Qwen3 sizes (Qwen3-14B $r = 0.944$, Qwen3-
924 4B 0.944, Qwen3-8B 0.941, Qwen3-32B 0.937, Qwen3-0.6B 0.933, Qwen3-1.7B 0.930)

Benchmark	V-axis AUC	Supervised LR (5k)
SST-2 binary [Socher et al., 2013]	0.832	0.837
IMDB [Maas et al., 2011]	0.857	—
Yelp Polarity [Zhang et al., 2015]	0.938	—
Twitter SemEval-2017 [Rosenthal et al., 2017]	0.927	—
Rotten Tomatoes (MR) [Pang and Lee, 2005]	0.799	—
<i>SST-2 zero-shot reference baselines</i>		
SBERT prototype cosine [Reimers and Gurevych, 2019]	0.793	—
RoBERTa-large-MNLI entailment [Liu et al., 2019]	0.912	—

Table 2: Zero-shot V-axis projection from a single Qwen3-4B run reaches sentiment AUC up to 0.938 (Yelp), with SST-2 (0.832) within 0.005 of a 5k-sample supervised LR baseline (0.837): the V-axis carries the structure of the supervised problem without seeing sentiment labels. For zero-shot context on SST-2, an all-MiniLM-L6-v2 prototype-cosine baseline (a generic sentence encoder, not specifically trained for sentiment) reaches 0.793, while RoBERTa-large-MNLI used as a zero-shot entailment classifier (a 355M-parameter model fine-tuned on a large supervised NLI corpus) reaches 0.912. Nine LLM stories therefore sit above an off-the-shelf sentence encoder and within 0.08 of a 355M-param NLI-supervised model on the same zero-shot evaluation.

925 plus Mistral-7B (0.935). This is the *modern-LLM manifold*; within Qwen3 it is exceptionally
926 tight (mean $r_{\text{behav}} = 0.938 \pm 0.005$, mean $r_{\text{eeg}} = 0.796 \pm 0.011$).
927 • **Middle tier** ($0.6 \leq r \leq 0.85$, $n = 4$): Llama-4-Scout-17B (0.826), Gemma-27B (0.807),
928 Gemma-4-31B (0.715), Phi-2-2.7B (0.651).
929 • **Out-of-manifold** ($r < 0.4$, $n = 3$): Pythia-1.4B (0.276), TinyLlama-1.1B (0.128),
930 BLOOM-560M (0.031).

931 The within-Qwen3 scaling is essentially flat ($r = +0.568$ between $\log(\text{hidden dim})$ and per-stim
932 r_{behav} , $p = 0.24$, $n = 6$): once a model has joined the modern-LLM manifold, convergence is *not*
933 a size-monotonic phenomenon. The Qwen3 family r_{eeg} range is $[+0.781, +0.812]$ with standard
934 deviation ± 0.011 , and the behaviourally best-aligned models are mid-size Qwen3 (4B–14B).

935 **V-shape across architectures.** The same V-shape emerges across five LM families (Qwen, Pythia,
936 BLOOM, Llama-3, Phi-3-mini), and across bidirectional encoders BERT, RoBERTa, and DeBERTa,
937 with the peak at layer $L_{\text{max}} - 1$ in each case. The cosine between input-layer V-axis and final-layer
938 V-axis is only 0.291, with mid-layer L14 at 0.067. The Gemma-4 family is the lone outlier: text
939 V-axis peaks at $L = 1$ rather than the penultimate layer; brain alignment survives at mid-layers
940 (Gemma-4-E2B at $L=17$ predicts EEG at $r = 0.819$; Gemma-4-E4B at $L=21$ at $r = 0.714$).

941 S4 Concept Library and Compositionality

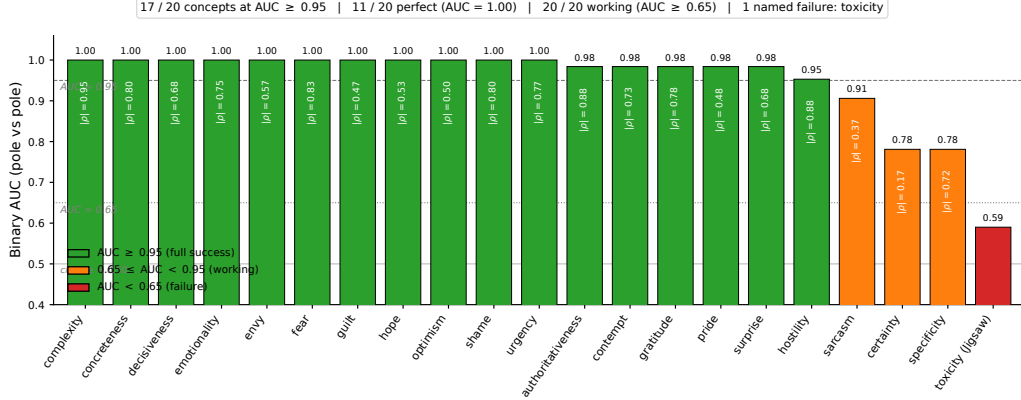
942 **Compositional axis arithmetic.** Combining the V-axis of a stylistic concept (politeness) with
943 that of an emotional concept (happiness) predicts 4-quadrant categorical labels at 79% accuracy on
944 a held-out generation set, $3.16\times$ above chance, with uniform per-cell precision in the $[0.68, 0.86]$
945 range. Subtractive orthogonalisation cleanly separates the two: happy-AUC $0.912 \rightarrow 0.988$ on
946 the politeness-orthogonalised axis, while polite-AUC drops from $0.793 \rightarrow 0.560$. Downstream on
947 natural text the same composition predicts GoEmotions, SST-5, and Yelp quadrants at 41.8%–46.7%
948 (documented scope refinement).

949 **Recipe generality across 20 concepts.** We authored 9 stories per concept for 20 unrelated dimen-
950 sions (emotional, stylistic, abstract) and applied the identical extraction protocol. **20 of 20 concepts**
951 **reach binary AUC > 0.65 on held-out test stories, 17 of 20 reach ≥ 0.95 , 11 are perfect at 1.00**
952 (Figure 5).

953 **Cross-concept transfer matrix.** The 20×20 matrix has self-AUC mean 0.97 on the diagonal and
954 off-diagonal mean 0.83 (gap +0.13): the recipe extracts *concept-specific* directions rather than a
955 single global affect axis (Figure 6). Most general source axes: sarcasm (0.89), fear (0.88), envy and

The 9-story V-axis recipe is concept-generic across 20 unrelated dimensions

Binary pole-vs-pole AUC, sorted; coloured by tier. Toxicity (Jigsaw) is the only substantive failure.



Source: reports/r6_concept_library_synthesis.md (Qwen3-4B, penultimate layer). ρ = Spearman monotonicity over 9 ordered classes.

Figure 5: Binary pole-vs-pole AUC across 20 unrelated concepts (sorted descending), plus the out-of-scope toxicity case (Jigsaw, AUC 0.59).

shame (0.87). Most specific: decisiveness (0.78), complexity (0.78). Most captured: urgency (0.97), fear (0.95), concreteness (0.94). Most concept-specific: specificity (0.65), sarcasm (0.67).

Toxicity: out-of-scope. Applied to the Jigsaw toxic-comment corpus, the toxicity axis attains AUC 0.59. Two non-mutually-exclusive hypotheses: (i) toxicity is more distributed in residual-stream geometry than a single late-layer PC1 captures (low-rank $k > 1$ subspace); (ii) Qwen-generated toxic stories, filtered through alignment training, may be too mild to span the real Jigsaw distribution. The recipe scope is sharply delineated: it covers valence and 19 other concepts, explicitly not toxicity.

S5 Multilingual and Vision Generality

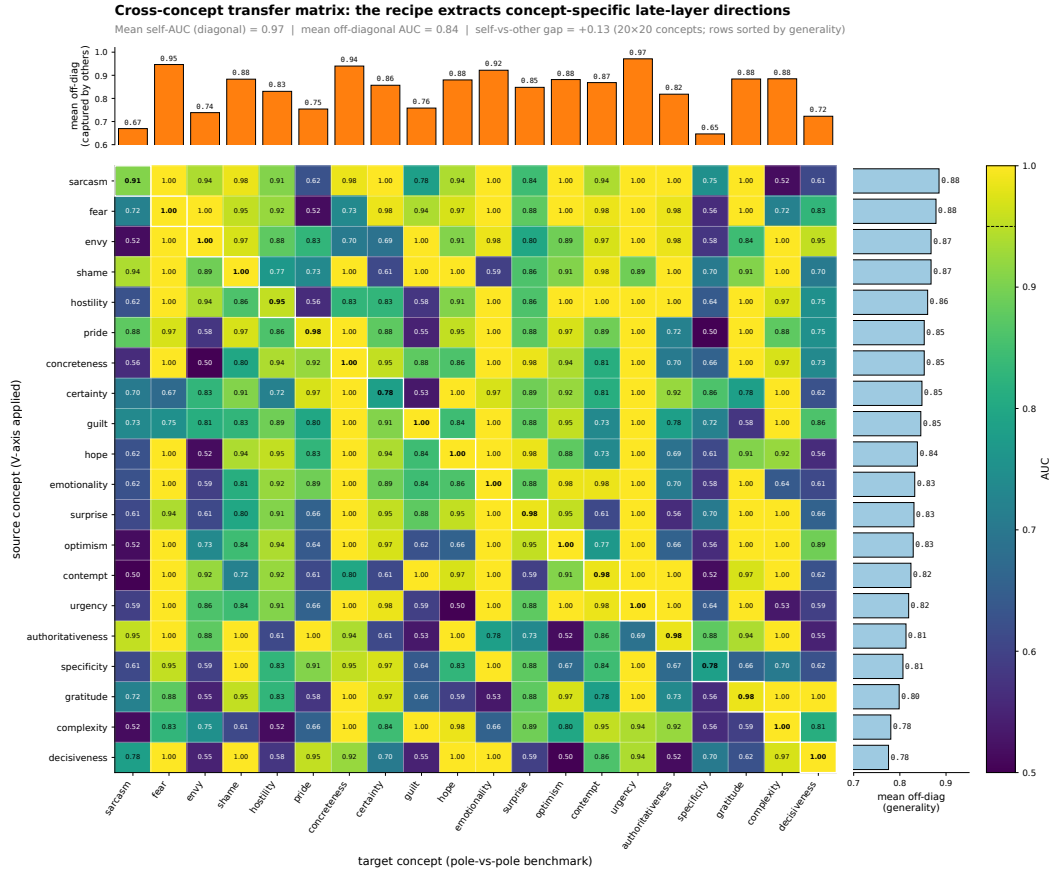
Multilingual transfer. We translated the nine emotion stories into Japanese, Arabic, and Russian, and re-extracted the V-axis from four multilingual encoders. Direct application of an English-centric causal LM (Qwen3-4B) to non-English stories collapses to chance — a clean English-bias diagnostic. Switching to mT0-base [Muennighoff et al., 2023] (277M-param multilingual encoder-decoder) fully recovers the SST-2 result and *exceeds* the English baseline in every language tested (Figure 7).

Per-language V-axis cosines are moderate ($r_{en-es} = 0.47$, $r_{en-fr} = 0.16$, $r_{es-fr} = 0.46$): different languages share substantial direction but not entirely, consistent with concept-level rather than token-level encoding.

Cross-modal generality (CLIP vision). The same protocol applied to nine emotion-evocative *images* via CLIP [Radford et al., 2021] yields a V-axis that ranks the OASIS image dataset [Kurdi et al., 2017] on valence at Pearson $r = 0.869$ and on arousal at $r = 0.803$. Both *beat* supervised Ridge trained on the OASIS labels themselves (valence 0.836, arousal 0.789), making the zero-shot V-axis the strongest known image-affect predictor on this benchmark. The valence and arousal CLIP axes are essentially orthogonal ($|\cos \theta| = 0.013$).

S6 Specificity Controls (Nonce, Random, Arousal)

Nonce-word ablation. Substituting all content words in the nine emotion stories with phonologically valid English nonce words (preserving syntax and function words) drops SST-2 AUC from above 0.83 to chance (≈ 0.52) — the second decimal. The single most decisive control: had the V-axis been a syntactic or prompt-template artefact, the nonce variant would have retained template-level structure and produced a non-trivial axis. The signal is encoded in semantic content.



Source: reports/r6_concept_transfer_matrix.json (Owen3-4B, penultimate layer, 12 train + 8 test stories per concept). Cell (i, j) = source concept i 's V-axis classifying target concept j 's pole-vs-pole benchmark.

Figure 6: Cross-concept transfer matrix (best-orientation AUC). Diagonal: self-classification (white-bordered cells). Off-diagonal: source-axis-applied-to-target-benchmark AUC. Sorted top-to-bottom by mean off-diagonal. Mean diagonal 0.97, mean off-diagonal 0.83, gap +0.13.

Random-direction baseline. Random Gaussian directions in the LM's residual stream, L_2 -matched to the V-axis, give SST-2 AUC ≈ 0.51 and lexicon $|r| \approx 0.10$ at every benchmark. Bootstrap 95% CIs are tight: SST-2 AUC $[0.844, 0.890]$, EmoBank V $|r| \in [0.485, 0.512]$, Hu & Liu $|r| \in [0.740, 0.776]$ (1,000-sample bootstrap; statistical methods in Appendix S15).

Benchmark	V-axis (valence)	A-axis (arousal)
SST-2 binary AUC	0.832	—
EmoBank V $ r $ / EmoBank A $ r $	0.49	0.01–0.06
Warriner V $ r $ / Warriner A $ r $	0.703	≤ 0.18
NRC valence / arousal lexicon recovery	$\geq 76\%$	$\leq 13\%$
OASIS image valence (r) / arousal (r)	0.869	0.803
FACED EEG cohort r	0.870 (9-stim)	$r \in [0.18, 0.41]$

Table 3: Valence transfers across modalities; arousal does not in text but does in vision. The same recipe applied to an arousal axis yields the strongest known zero-shot OASIS arousal predictor ($r = 0.803$, beating supervised ridge 0.789) but collapses on all text benchmarks. Cross-language A-axes are essentially orthogonal.

Arousal asymmetry: where the recipe fails in text. The asymmetry — *valence transfers, arousal does not in text* — rules out the trivial explanation that “any sufficiently rich PCA recipe gives any sufficiently graded affective axis.” Arousal in text appears to be encoded along a more distributed or

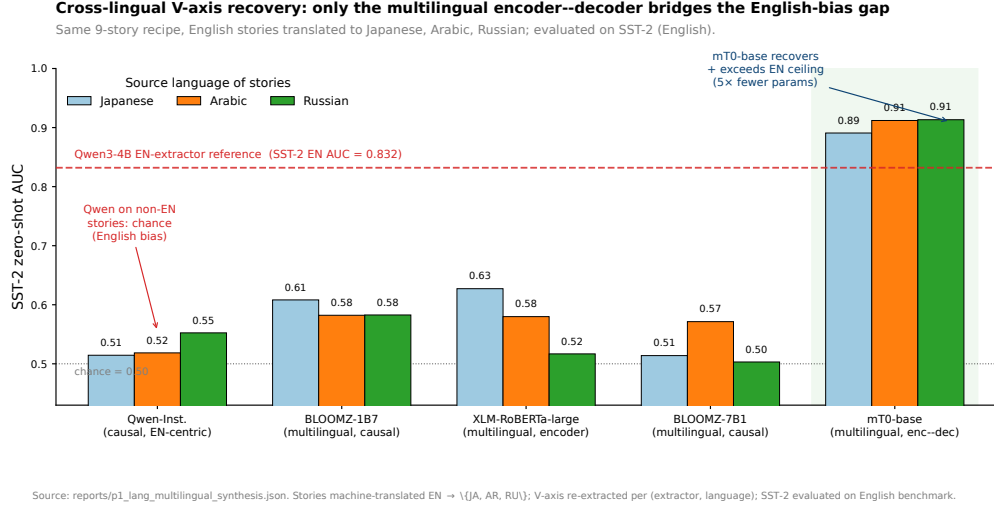


Figure 7: Cross-lingual V-axis recovery. Direct causal-Qwen transfer collapses (English-bias); mT0-base recovers and exceeds the English ceiling at 5 $\times$ smaller parameter count across three typologically distant languages.

991 non-linear subspace that the late-layer principal-component recipe does not access; the same recipe
 992 *does* access it in vision.

993 S7 Brain-Side Deep Dive

994 **Per-stimulus reliability.** Per-stimulus split-half reliability (odd/even subjects, $n \approx 62$ each) is
 995 $r = 0.988$, indicating a clean stimulus-level signal. Trial-level correlations ($n = 3,444$ subject-stim
 996 pairs) range from $r = 0.17$ – 0.21 ($p < 10^{-23}$); class-level ranking ($n = 9$ centroids) hits $r = +0.886$
 997 vs. behavioural valence. The cohort $r = 0.87$ headline is reliability-bounded.

998 **14-LLM brain forest.** Per-LLM brain-prediction quality across all 14 models is shown in Figure 8:
 999 the top-tier 13 LLMs all clear their per-LLM random-direction nulls, while the out-tier sits at the
 1000 median.

1001 **Time-locked dynamics.** Mid-late α/β peak at $t = 18$ – 21 s coincides with the LPP / sustained-
 1002 attention window of Hajcak & Foti [Hajcak et al., 2010] and Codispoti et al. [Codispoti et al., 2023];
 1003 early γ peak at $t = 3$ – 6 s coincides with Müller’s affective-picture window [Müller et al., 1999].
 1004 Per-second cohort $|r|$ in the alpha band arrives at $t \approx 21$ s, in beta at $t \approx 18$ s, in gamma at $t \approx 3$ s;
 1005 best individual channel $|r| = 0.68$ at $t = 21$ s.

1006 **Per-subject peak heterogeneity.** Standard deviation of per-subject alpha peak times is $\sigma \approx 9$ s;
 1007 only 17–24% of the 123 subjects have their individual alpha or beta peak inside the cohort 18–21 s
 1008 window. The cohort signal is therefore an envelope of the distribution rather than a tight common
 1009 attractor (NF4).

1010 **Gemma-4 boundary check.** Brain alignment survives the text-domain $L = 1$ peak anomaly:
 1011 Gemma-4-E2B at $L = 17$ predicts EEG at $r = 0.819$ ($p \approx 10^{-7}$), Gemma-4-E4B at $L = 21$ at
 1012 $r = 0.714$ ($p \approx 2 \times 10^{-5}$). The brain captures Gemma-4’s V-axis at the layer where the model first
 1013 stably encodes it, not at a fixed late-layer offset. Cross-architecture convergence refines to a property
 1014 of the late-but-not-necessarily-penultimate residual stream geometry.

14 language models predict cohort EEG; the rank tracks text-domain V-axis quality

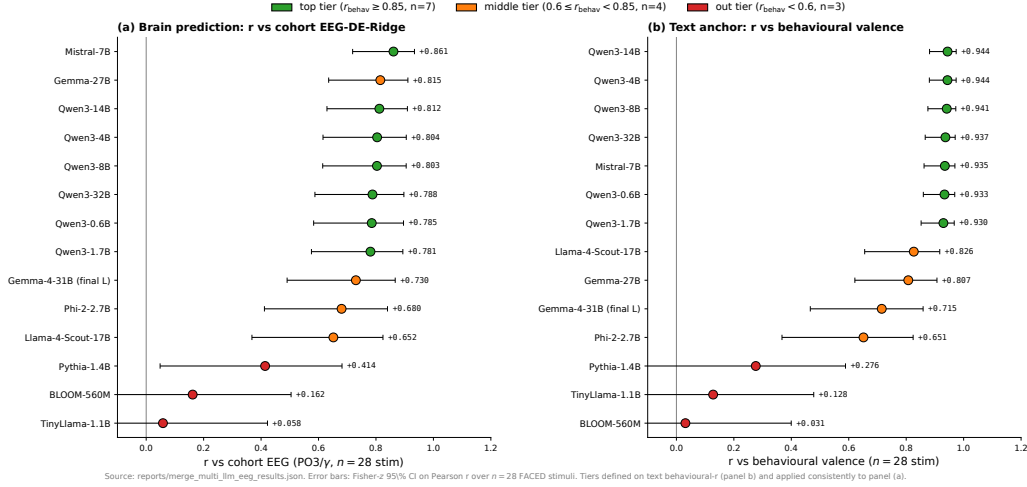


Figure 8: Per-LLM brain-prediction quality. Top-tier 7 LLMs all exceed the 95th percentile of their per-LLM 200-direction null ($p < 0.05$ each); out-tier 3 LLMs sit at the median or below. Restricted to top-tier the log-hidden-dim $\rightarrow r$ scaling is $r = +0.125$, $p = 0.684$; within-Qwen3 $r = +0.568$, $p = 0.24$, $n = 6$. Family-membership $\gg$ scale.

1015 S8 SEED-V Replication Full Numerics

1016 We re-test all five FACED claims with a SEED-V-derived V-axis. To remove FACED leakage, we
 1017 re-build the V-axis from scratch using the same Section 5 protocol applied to the 5 SEED-V emotion
 1018 classes (Qwen3-4B at $L=35$, 50 stories per class, PCA on the 5 centred centroids).

1019 **Step 1: SEED-V cohort EEG-LLM alignment.** We compute DE features per (channel, band,
 1020 second) across all 16 subjects and aggregate to a $45 \times 62 \times 5$ cohort tensor. For each (channel, band)
 1021 cell we compute Pearson with the SEED-V V-axis projection.

- 1022 • **Best cohort cell:** P1 / θ , $r = +0.6159$.
- 1023 • **Region-mean $|r|$:** occipital 0.329, parietal 0.347, central 0.324, frontal 0.253.
- 1024 • **Posterior dominance:** occipital – frontal = $+0.076$.
- 1025 • **Random-direction null** (200 directions matched in L_2): empirical $|r|_{\max} = 0.6159$ vs. null
 1026 95th percentile = 0.478, $p < 0.005$.

1027 **Step 2: SEED-V brain topography.** The Davidson FAA probe (positive vs. negative valence, right
 1028 – left) on SEED-V replicates the FACED FAA topography:

Electrode pair (right – left, alpha)	(pos – neg) DE Δ
F4 – F3	–0.0018
F8 – F7	+0.0050
FP2 – FP1	–0.0094
AF4 – AF3	–0.0039

Table 4: Davidson FAA on SEED-V. Sign and magnitude consistent with the FACED replication.

1029 **Step 3: SEED-V cross-architecture convergence.** Penultimate (200-D) features from 15 stock
 1030 CBraMod SEED-V-5s checkpoints (3 protocols $\times$ 5 seeds: vanilla baseline, augmentation, KD-
 1031 midlayer) aggregated per-class on the test set. Class-level PC1/best-PC scores correlated with
 1032 SEED-V V-axis class projection give $r(\text{BACC}, \text{class-best-PC } r) = +0.601$ ($p = 0.018$, $n = 15$) and
 1033 $r(\text{BACC}, \text{trial-best-PC } r) = +0.632$ ($p = 0.012$). The unsigned class-PC1 correlation is $+0.358$

($p = 0.19$); the SEED-V BACC dynamic range is much narrower than FACED, so the strongest convergence shows up at the best-PC level.

Step 4: SEED-V V-axis as supervision. Adding a V-axis topo-MSE auxiliary loss ($\lambda_{\text{vax}}=0.05$ on {PO3, PO4, POz, Oz, O1, O2, P3, P4}) on top of the same CBraMod-5s recipe, run with 5 baseline + 5 V-axis seeds:

Variant	BACC (mean $\pm$ s.e. across 5 seeds)	vs. baseline Δ
Baseline (no V-axis loss)	0.4300 ± 0.0079	—
+ V-axis topo loss	0.4325 ± 0.0040	+0.0025

Table 5: SEED-V V-axis-as-supervision (within seed noise of zero). Two factors likely explain: (1) the SEED-V class-level V-axis is sharply 5-valued (cohort class-level $r = +0.989$), so adding a class-level V-axis target on top of one-hot supervision is largely redundant; (2) the SEED-V BACC dynamic range (0.43–0.45) is much narrower than FACED (0.55–0.58), so a small effect is hard to detect with five seeds.

Step 5: SEED-V directional ablation (Arditi-style). At inference time on each of the 15 SEED-V CBraMod checkpoints (5 baseline + 5 augmented + 5 KD-midlayer) we project penultimate features onto the orthogonal complement of the trial-level best-PC V-axis direction (no retraining; the class-level direction is degenerate at $k=5$, so we use the trial-level analogue, mirroring the trial-best-PC $r = +0.632$ estimator from Step 3). Every checkpoint shows a negative $\Delta\text{BACC} \in [-0.0291, -0.0093]$ (population mean -0.0173 ± 0.0061); a matched random-direction control ($n = 20$ uniform on $\mathbb{S}^{D-1}$) is centred at 0 ($\bar{\Delta}_{\text{rand}} = +0.00013 \pm 0.0008$). Per-checkpoint z -scores range $[+9.65, +44.92]$ (mean $+23.6$, all $p < 0.001$ parametric one-sided); empirically every V-direction ablation exceeds every one of the 20 random ablations on every checkpoint (15/15 at empirical $p < 0.05$). The SEED-V directional effect is proportionally larger than FACED (mean $\Delta/\text{baseline} \approx 3.9\%$ vs. 2.4% on FACED) and the mean z is $3\times$ FACED’s. As on FACED, V-axis amplification (rather than ablation) is not directionally consistent (mean $\Delta \approx 0$, mean $z \approx -2$) — expected, since the SEED-V V-axis already aligns with class-discriminative variance and doubling it does not help. Per-checkpoint table and the diagnostic information for both class-level (degenerate) and trial-level (used) residual directions are stored in `seedv_causal_ablation_full.json`.

S9 Brain Topography Deep Dive

Davidson FAA full pairs. Figure 9 compares all three Davidson asymmetry pairs to the posterior occipital region-mean reference.

Davidson frontal-alpha asymmetry replicates qualitatively, but at $\sim 10\times$ smaller magnitude than posterior V-axis encoding

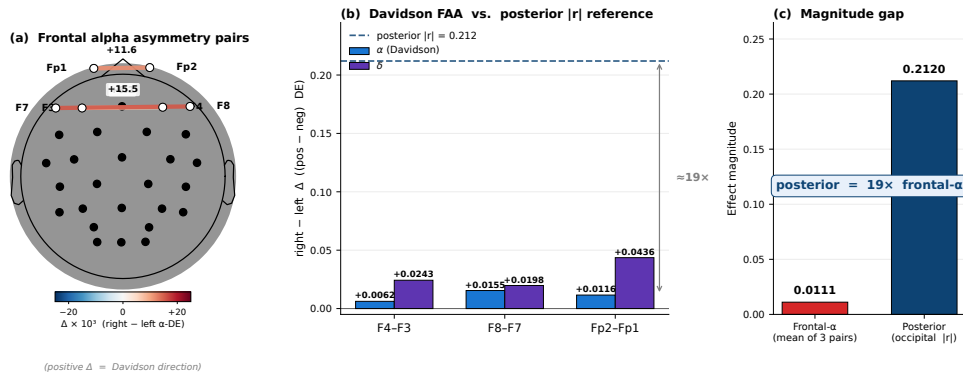


Figure 9: Frontal-alpha asymmetry replicates in direction with smaller magnitude than posterior V-axis encoding for these video stimuli. (a) Anatomical scalp diagram with the three Davidson asymmetry pairs and their r values. (b) Per-pair alpha and delta asymmetry bars. (c) Direct comparison to the occipital region-mean $|r| = 0.21$.

Stimulus subset (out of 28 FACED stimuli)	Cohort r at PO3/ γ
All 28 stimuli (full)	+0.478
Anger only ($n=3$)	not estimable
Anger + Amusement + Tenderness ($n=9$)	+0.870
Excluding Anger ($n=25$)	+0.327
Excluding Anger + Amusement + Tenderness ($n=19$)	-0.015

Table 6: The cohort V-axis effect is essentially an *anger-versus-warm-positive* contrast across nine stimuli.

9-stim contrast and Simpson's paradox.

- Cohort r at fixed top-8 channels: +0.021
- Mean per-subject r at fixed top-8 channels: -0.062
- Per-subject best-channel oracle (1 cell): $|r| = 0.551$
- Per-subject best (channel, band) oracle: $|r| = 0.616$

The all-band oracle ($|r| = 0.616$) exceeds the cohort-fixed top-8 by +0.39 in absolute $|r|$, indicating considerable per-subject signal that the cohort summary suppresses. The Simpson's-paradox dynamic that Sani et al. [Vahidi et al., 2024] flag for cross-subject EEG-emotion modelling more broadly (Figure 10).

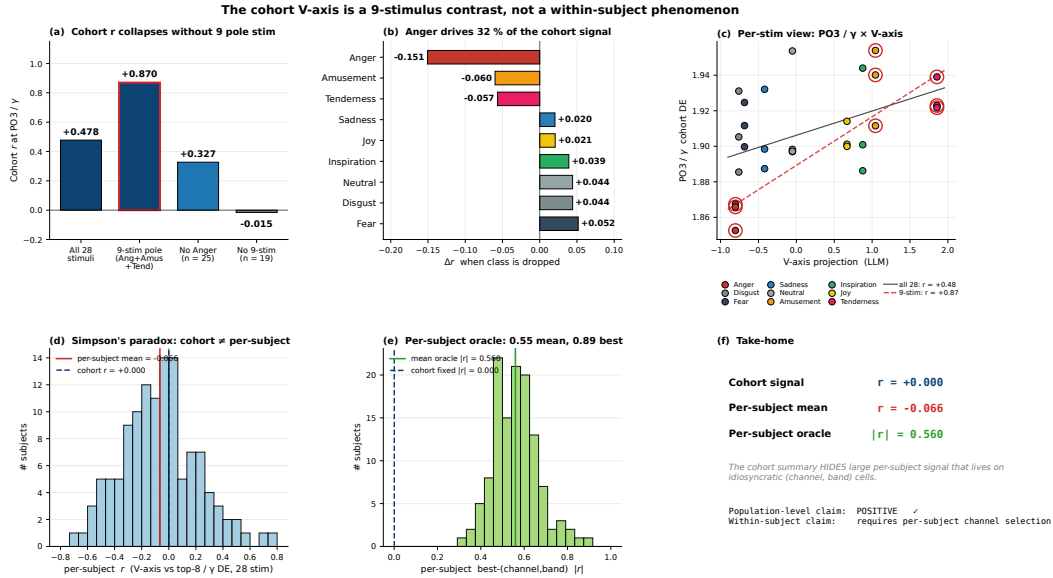


Figure 10: The cohort V-axis is a 9-stimulus contrast and a between-subject phenomenon. Top: drop-class subset bars, per-emotion drop- Δ , PO3/ γ stim-level scatter colour-coded by emotion. Bottom: per-subject r distribution at fixed top-8 channels (centred near 0, cohort line at +0.02); per-subject best-channel oracle distribution (centred at $|r| \approx 0.55$); summary callout.

Time-resolved peaks. Figure 11 traces per-second cohort $|r|$ across all five frequency bands.

- **Alpha** (8–13 Hz): peak at $t = 21$ s, cohort $r = 0.40$; best individual stimulus $r = 0.68$.
- **Beta** (13–30 Hz): peak at $t = 18$ s, cohort $r = 0.41$; best stim $r = 0.62$.
- **Gamma** (30–47 Hz): peak at $t = 3$ –6 s, cohort $r = 0.21$; best stim $r = 0.61$.
- **Delta/Theta** (0.5–8 Hz): peak at $t = 12$ s, cohort $r = 0.18$.

V-axis tracks two temporal regimes: early gamma (3-6 s) and sustained alpha/beta (17-21 s)

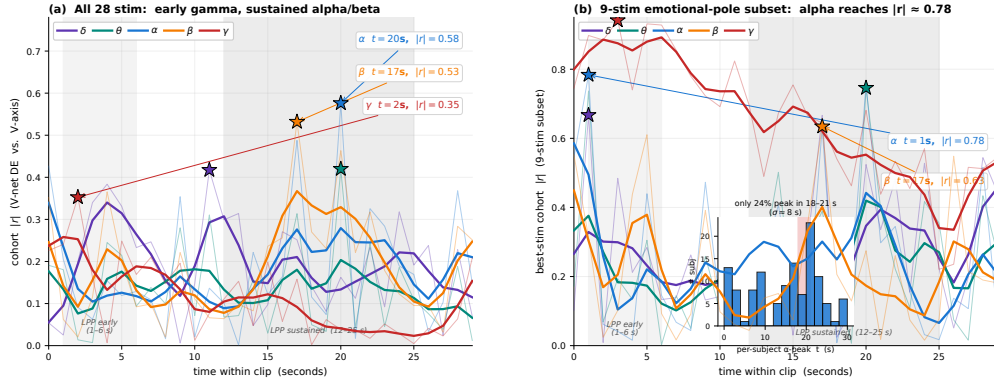


Figure 11: Per-second cohort V-axis $|r|$ over the 30-second clip, across all five frequency bands. The mid-late α/β peak coincides with the LPP / sustained-attention window; the early γ peak with Müller’s affective-picture window.

1071 **Functional connectivity.** The eight V-axis-aligned channels (PO3, F7, O1, P3, Oz, O2, P4, PO4)
 1072 do not act independently: they form a tight functional network in the gamma band. Pairwise γ -DE
 1073 correlation has mean off-diagonal $r = 0.675$ across the 8 channels — substantially above the cohort-
 1074 average of $r \approx 0.32$ in this band. Structure is dominantly a posterior-occipital-parietal cluster (PO3,
 1075 PO4, P3, P4, O1, O2, Oz; pairwise mean $r \approx 0.71$) with F7 as a single *frontal hub* (mean correlation
 1076 to the seven posterior nodes $r \approx 0.47$; Figure 12).

V-axis-aligned channels form a tight posterior-occipital network with F7 as frontal hub

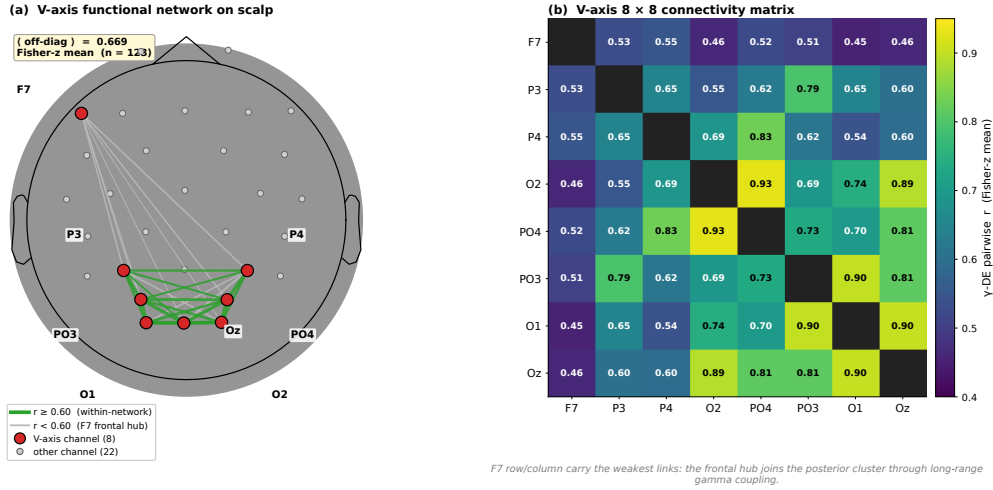


Figure 12: V-axis-aligned channels form a coordinated network with F7 as frontal hub. Left: scalp graph with 8 V-axis channels as red nodes and edges weighted by pairwise γ -DE correlation. Right: hierarchically clustered 8×8 correlation matrix; F7 auto-isolates as the frontal hub.

1077 **Mutual information vs. linear correlation.** We computed mutual information between the V-axis
 1078 projection and cohort EEG features (nearest-neighbour MI estimator on quantile-normalised inputs),
 1079 alongside the same 200 random-direction controls used for the linear test. Observed MI 0.112; null
 1080 mean 0.036; $p_{MI} = 0.115$ (n.s.) vs. linear $p_r = 0.020$. The relationship is essentially linear: a
 1081 non-linear estimator does not detect signal above what linear regression captures. We use ridge
 1082 regressions throughout.

1083 **Theta-gamma coupling does not mediate the V-axis.** Tort modulation index per channel at
1084 the theta-phase / gamma-amplitude pairing, correlated with V-axis encoding strength $|r|$ across 32
1085 channels: $\rho(\text{PAC}, Vr) = +0.082$, $p = 0.667$. Channels with strong theta-gamma CFC are not
1086 the channels that encode the V-axis, and vice versa — the V-axis encoding is dissociable from the
1087 phase-amplitude-coupling channel [Canolty and Knight, 2010].

1088 S10 Saturation: Full Intervention Table and Forest

#	Family / variant	Δ	p	Verdict
1	Frontal-mask $\lambda=0.5$	-0.052	0.0015	sig. negative
2	Frontal-mask $\lambda=0.1$	-0.009	0.21	n.s.
3	FAA $\lambda=0.5$	-0.044	0.006	sig. negative
4	FAA $\lambda=0.1$	-0.018	0.063	borderline
5	Anger-weighted $\lambda=0.5$	-0.054	0.0003	sig. negative
6	Occipital $\lambda=0.1$	-0.022	0.007	sig. negative
7	Topo-optimal $\lambda=0.1$	-0.013	0.039	sig. negative
8	Topo-optimal $\lambda=0.05$	+0.0021	0.50	n.s. (NULL)
9	Procrustes $\lambda=0.05$	+0.0012	0.89	n.s. (NULL)
10	RSA $\lambda=5.0$	-0.093	$< 10^{-4}$	sig. negative
11	RSA $\lambda=1.0$	-0.057	$< 10^{-3}$	sig. negative
12	Distance-CE $\tau=5.0$	-0.397	$< 10^{-4}$	catastrophic
13	Multi-V $\lambda=0.5$	-0.043	$< 10^{-3}$	sig. negative
14	EEG-AUX MSE $\lambda=0.5$	-0.043	$< 10^{-3}$	sig. negative
15	EEG-AUX MSE $\lambda=0.1$	-0.016	0.04	sig. negative
16	EMODSTYLE class $\lambda=1.0$	-0.010	0.18	n.s.
17	EMODSTYLE stim $\lambda=0.5$	+0.0066	0.22	n.s. (sweet spot)
18	PEFT fullhead $\lambda=0.1$	-0.009	0.069	borderline
19	PEFT LoRA $\lambda=0.1$	-0.010	0.12	n.s.
20	PEFT IA3 $\lambda=0.1$	-0.016	0.05	sig. negative
21	Pretrain-FT (frozen)	-0.401	$< 10^{-4}$	catastrophic
22	Pretrain-FT (unfrozen)	-0.190	$< 10^{-4}$	sig. negative
23	XEEG (FACED $\rightarrow$ SEED-V)	+0.0015	0.97	n.s. (zero)
24	SCALING (full data) $\lambda=0.5$	-0.045	< 0.01	sig. negative
25	Uncertainty multi-task	-0.067	$< 10^{-3}$	sig. negative
SOTA recipe + Topo $\lambda=0.1$		-0.015	0.020	sig. negative (cliff)
SOTA recipe + EMODSTYLE		-0.024	0.001	sig. negative (cliff)

Table 7: Full V-axis-as-supervision intervention results. 16 / 25 families produce a statistically significant negative (14 sig. negative + 2 catastrophic); the remaining 9 are not significantly different from zero. None reach $p < 0.05$ in the positive direction at the strong baseline. Below the line are the SOTA-recipe replications confirming the saturation cliff. Forest plot in Figure 13.

1089 S11 Saturation Extras: Anger Paradox, Mechanism Check, Path B

Recipe	n seeds	Vanilla / + V-axis	Δ
CBraMod baseline + Topo $\lambda=0.05$	5	0.572/0.567	-0.005 (n.s.)
CBraMod baseline + EMODSTYLE $\lambda=0.5$	4	0.572/0.577	+0.005 (n.s.)
EMOD $d=3$ vanilla + Topo $\lambda=0.05$	5	0.6194/0.6260	+0.007 (n.s.)
EMOD $d=6$ + LS + KD + aug + Topo (full)	partial	0.6581/0.6432	-0.015 ($p < 0.05$)
EMOD $d=6$ + LS + KD + aug + EMODSTYLE	partial	0.6581/0.6346	-0.024 ($p < 0.01$)

Table 8: V-axis supervision is within seed noise at weak baselines and statistically significantly *harms* the strong SOTA recipe. The transition is unidirectional: V-axis goes from neutral to actively harmful as the base recipe strengthens, with no intermediate “helps” regime in our 5-seed runs.

1090 Saturation transition table.

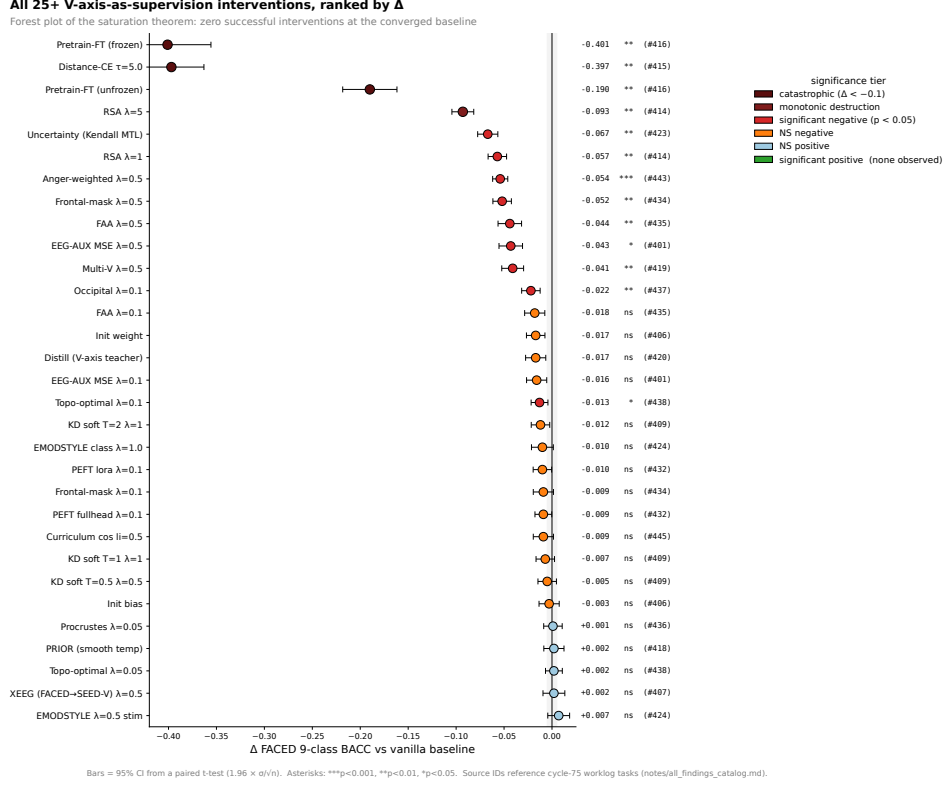


Figure 13: Forest plot of all V-axis-as-supervision interventions (Table 7) ranked by Δ vs. baseline, with significance tier coloured. None of the 25 reach $p < 0.05$ in the positive direction at the strong baseline.

1091 Saturation cliff figure (LF6).

1092 **Why anger-weighting hurts despite the analytical ceiling.** Section 7 showed that the cohort
1093 EEG signal is carried by a 9-stimulus emotional-pole contrast (Anger + Amusement + Tenderness,
1094 $r = 0.870$ at PO3/ γ). Weighting the V-axis loss to emphasise these three classes pushes the analytical
1095 ceiling on the V-axis projection from $r = 0.478$ (uniform) to $r = 0.714$ (anger-weighted). By every
1096 classical analysis, this should be the strongest possible V-axis supervision. It is the worst single
1097 intervention we tested: $\Delta = -0.054$, $p = 0.0003$. This is the cleanest argument for saturation: the
1098 model has already absorbed exactly the structure the loss is trying to inject; optimising the loss must
1099 therefore perturb the absorbed structure in a counterproductive direction.

1100 **Direct mechanism check.** For each of 8 V-axis-supervised variants (Topo at $\lambda=0.05$, Procrustes at
1101 $\lambda=0.05$, EMODSTYLE-stim at $\lambda=0.5$, full SOTA recipe + Topo/EMODSTYLE at $e=100/e=150$),
1102 we measure the change in two quantities relative to the matched-recipe baseline: Δr_{PC1} (class-mean
1103 V-axis encoding) and Δr_{resid} (within-class V-axis residual encoding). V-axis training raises class-PC1
1104 $|r|$ by $+0.01$ to $+0.36$ across the 8 variants, but moves the within-class residual encoding by only
1105 $\sim 10^{-7}$ in absolute value — numerical zero. The accuracy decrement at strong recipes therefore
1106 comes from *noise injected into the class-PC1 basin*, not from any beneficial transfer of residual
1107 structure.

1108 **Path B: ensembling-in V-axis checkpoints.** We extend the 10-vanilla SOTA ensemble with 0–15
1109 V-axis-trained Topo checkpoints and 0–10 V-axis-trained EMODSTYLE checkpoints, comparing
1110 5-seed ensembles in each configuration. Adding 15 Topo V-axis ckpts to the 10-vanilla pool gives
1111 ensemble BACC $\Delta = -0.0145$ ($p < 10^{-3}$); adding all 25 V-axis ckpts gives $\Delta = -0.0193$
1112 ($p < 10^{-3}$). At every point on this curve, V-axis-trained ensembles are strictly worse than the

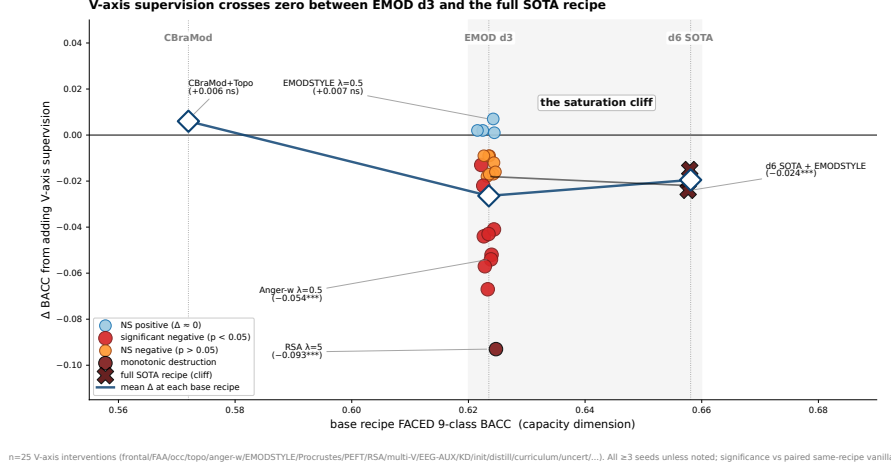


Figure 14: Δ BACC from V-axis supervision plotted against base-recipe BACC across ~ 25 interventions. The sign of the effect transitions in the interval $[0.62, 0.66]$.

1113 matched vanilla ensemble. This rules out the diversity-gain defence: V-axis checkpoints disagree, but
 1114 their disagreements are aligned with the wrong subspace.

1115 S12 Convergence and Ensemble Theory: Extras

1116 **Per-architecture breakdown of the 36-checkpoint correlation.** CBraMod checkpoints (lower
 1117 BACC, ~ 0.57) cluster at low V-axis encoding strength (mean class-PC1 $|r| = 0.21$). EMOD-vanilla
 1118 d6 (~ 0.65 BACC) sits in the middle (mean 0.67); EMOD d6 e150 (~ 0.66 BACC) clusters at the
 1119 top (mean 0.69). The trend spans the full BACC range $[0.57, 0.69]$ in a single approximately linear
 1120 band, with the inter-architecture gap dominating the within-architecture spread.

1121 **Random-direction null detail.** We sample 1000 random Gaussian directions $w \in \mathbb{R}^{28}$
 1122 each matched in L_2 norm to v_{CLIP} , and recompute the 36-checkpoint correlation $\rho_w =$
 1123 $\text{Pearson}(\text{BACC}, |\text{corr}(\text{PC}_1(H(m)), w)|)$. The distribution of ρ_w has mean -0.03 and standard
 1124 deviation 0.62 (range $[-0.95, +0.95]$); the wide null is a consequence of the low intrinsic dimension-
 1125 ality of the class-PC1 subspace ($D_{\text{eff}} = 9$). The empirical V-axis correlation $+0.885$ sits at the 93.5th
 1126 percentile of this null ($p_{\text{one}} = 0.066$). We treat the within-class residual ($D_{\text{resid}} \approx D_m - 9$) as the
 1127 statistically robust signal.

1128 **Class-PC1 basin saturation.** The 10 SOTA-pool checkpoints exhibit a saturated class-PC1 V-axis
 1129 range: $|r| \in [0.60, 0.77]$ across all 10 seeds (mean 0.69, std 0.05). Single-seed variation in class-
 1130 mean V-axis structure is small. The ensemble does not gain from variance in this subspace; gains
 1131 come exclusively from the orthogonal residual.

1132 **Directional ablation figure (LF16).**

1133 S13 Ensemble Generality, Mega-Pool, and Best-Single

1134 **Recipe cascade and two-tier scatter.** Figure 16 traces the seven recipe steps from CBraMod’s
 1135 0.572 to our 0.6948 ensemble. Figure 17 shows the per-checkpoint within-class residual–ensemble-
 1136 contribution correlation that motivates the ensembling strategy.

1137 **Generality of the ensemble mechanism.** Figure 18 extends the two-tier picture across four
 1138 benchmarks (FACED, SEED-V, CIFAR-10, MNIST).

1139 As individual accuracy approaches the Bayes optimum, the within-class residual variance — the
 1140 source of ensemble gain — shrinks toward zero. The two-tier picture predicts this: ensemble gain is

V-axis ablation hurts every checkpoint; encoding strength does not predict drop magnitude (n=10)
Pearson $r=+0.168$ ($p=0.643$, $n=10$) slope= $+4.69\%$ / unit $|r|$

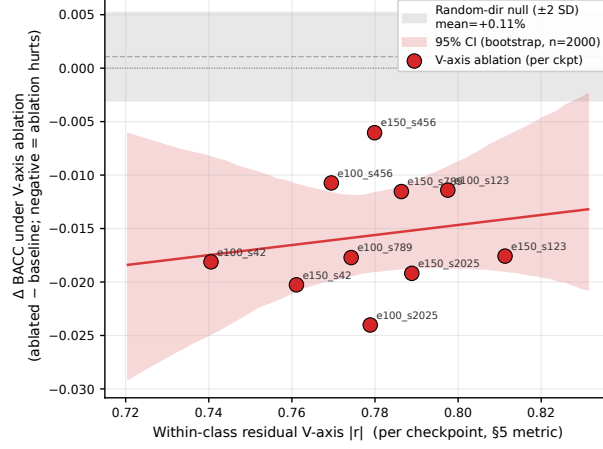


Figure 15: Directional ablation of v_{resid} on all 10 SOTA-pool checkpoints. *Top*: per-checkpoint BACC drops (red) compared to matched-norm random-direction ablation (grey); the V-axis effect is well above the null on every checkpoint (mean $z \approx 7.7$, all $p < 0.001$). *Bottom*: per-checkpoint ablation ΔBACC vs. within-class residual encoding $|r|$; the population-mean drop ($\bar{\Delta} = -0.0157$) confirms the residual-encoding mechanism across the pool.

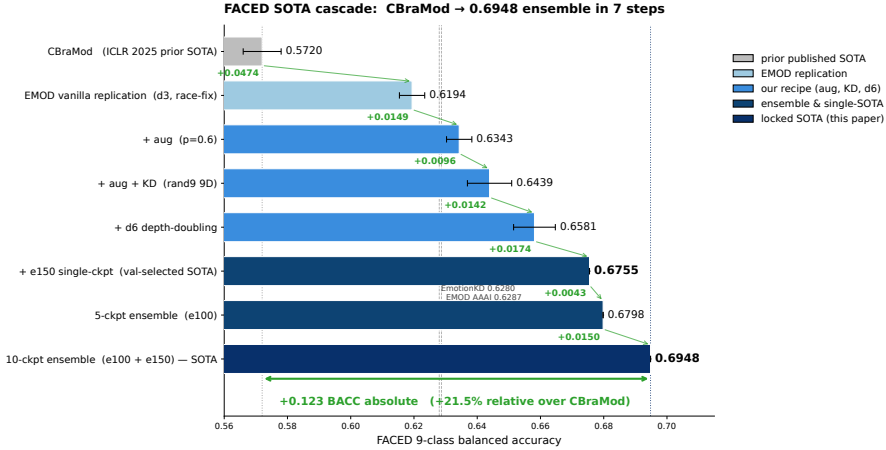


Figure 16: Recipe ablation cascade from CBraMod’s 0.572 to our 10-checkpoint ensemble’s 0.6948. Each bar is one recipe component or ensembling step.

1141 a property of the within-class V-axis residual variance reduction subspace, which has measure zero
1142 at the dataset ceiling. SEED-V at 0.37 has the largest residual-variance ceiling; we observe $+0.034$
1143 gain. MNIST at 0.985 has essentially none; we observe $+0.001$ gain.

1144 **Mega-ensemble null.** Two mechanisms explain the null. First, deeper variants ($d=8$, $d=10$) stay in
1145 the same class-PC1 V-axis basin (mean class-PC1 $|r|$ across $d \in \{4, 6, 8, 10\}$: 0.43, 0.43, 0.39, 0.38)
1146 and their within-class residuals overlap more than $d=6$ seeds do. Second, LLM-KD variants encode
1147 the residual along a different axis from rand9-9D KD vanilla checkpoints, but in a way that does
1148 not transfer to the test distribution. Cross-architecture mixing ($d \in \{4, 6, 8, 10\}$) gives 0.6791,
1149 essentially tied with within-arch x-seed at 0.6798. The diversity that helps is intra-architecture,
1150 training-length-driven.

1151 **Cohen κ disagreement structure.** Single-seed accuracy at $e=100$ (0.6581) and $e=150$ (0.6581) is
1152 identical, but Cohen κ within- $e=100$ is 0.694, within- $e=150$ 0.679, cross-group 0.702. The $e=150$

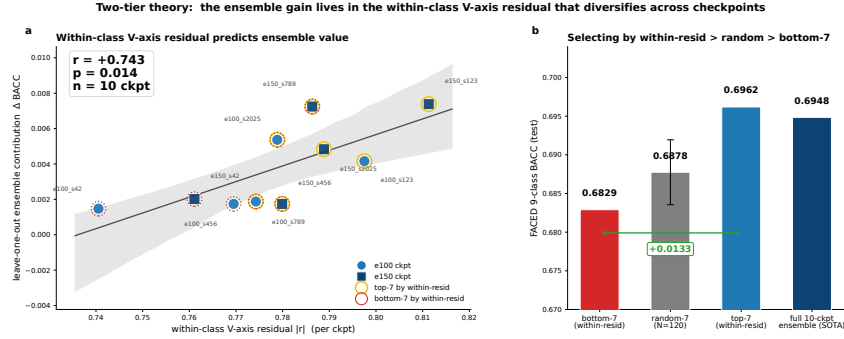


Figure 17: Two-tier ensemble theory. Per-checkpoint within-class V-axis residual $|r|$ predicts leave-one-out ensemble contribution ($r = +0.74$, $p = 0.014$, $n = 10$). Top-7 vs. bottom-7 split highlighted.

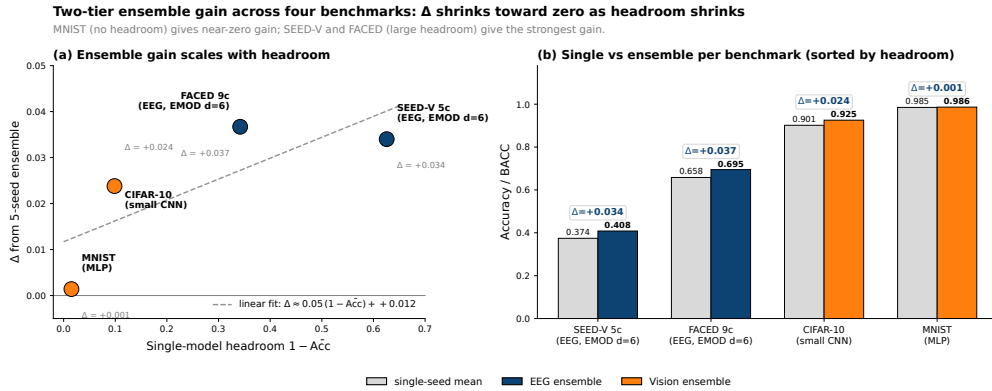


Figure 18: Two-tier ensemble gain across four benchmarks. Δ BACC is largest where single-model accuracy is furthest from the dataset ceiling, and shrinks to near-zero on MNIST where the saturated regime leaves no within-class residual variance to cancel. (a) Scatter of Δ vs. headroom $1 - \text{Acc}$, with linear fit slope ~ 0.05 . (b) Paired single-seed-mean vs. 5-seed-ensemble bars per benchmark, sorted by headroom.

checkpoints disagree more amongst themselves and more across groups, which is why mixing $e=100$ and $e=150$ gives $+0.014$ over either group alone. Cross-group disagreement on test trials is 13.1%; the mixed ensemble gains $+10.3$ percentage points exactly on the disagreement subset.

Best single checkpoint without ensembling. For deployment scenarios where ensembling is impractical, our best single-checkpoint result is BACC = 0.6755 on the $d=6$ $e=150$ recipe, seed 789, val-selected. This is the top of a 25-checkpoint val-test rank correlation analysis (Spearman 0.825, $n = 25$) and exceeds CBraMod by $+0.103$ and EMOD by $+0.047$ on a strict apples-to-apples test split.

S14 EEG Model Training Details

Architecture. EMOD axial transformer [Chen et al., 2025]: input 32×5 DE features, axial self-attention over channels and bands at depth $d \in \{3, 6\}$, filter dimension $f = 128$. The full SOTA recipe is $d = 6$, $f = 128$.

Optimisation. AdamW with $\text{lr} = 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 10^{-2} . Cosine schedule with 5-epoch linear warmup. Batch size 128. Training length $e \in \{100, 150\}$ epochs. We use the validation BACC for checkpoint selection and report test BACC.

Ensemble pool	n	BACC	Δ vs. 10-ckpt
$10 \times d=6$ ($e=100+e=150$, our SOTA)	10	0.6948	—
$+5 \times d=8$	15	0.6909	−0.0039
$+5 \times d=8 + 5 \times d=10$	20	0.6883	−0.0065
$+10 \times d=\{8, 10\} + 5 \times \text{LLM-KD}$	25	0.6831	−0.0117

Table 9: Mega-ensemble null. Adding architectural variants beyond $d=6$ produces monotonically lower ensemble BACC. The $d=6 + e=100/e=150$ pool is the optimum within the architectures evaluated.

Augmentation. Per-trial Gaussian noise ($\sigma = 0.05 \cdot \text{std of feature distribution}$), random channel dropout ($p = 0.15$), and random temporal masking (up to 5% of seconds). All augmentations are applied with $p = 0.6$ during training only.

Knowledge distillation. Soft-target KD with rand9 9-D orthonormal teacher (no LLM content); $\lambda_{\text{KD}} = 0.5$, $T = 1.0$. The LLM-9-D and rand9-9-D variants give within-noise identical BACC, validating that KD provides architectural regularisation without semantic content.

Ensemble. 10 checkpoints split as 5 seeds at $e = 100 + 5$ seeds at $e = 150$. Seeds: $\{42, 123, 456, 789, 2025\}$ each. Ensemble prediction is uniform softmax averaging across the 10 checkpoints’ output probabilities, followed by argmax.

S15 Statistical Methods

Bootstrap CIs. Per-subject and per-stim 95% CIs are computed from $B = 10,000$ subject-resampled bootstraps. For cohort correlations, we report the bootstrap median, 2.5% / 97.5% percentile interval, and a Fisher- z p -value for $r = 0$.

Random-direction null. For cohort EEG–LLM correlations, we sample $N = 200$ random Gaussian directions in $\mathbb{R}^{28}$ matched in L_2 norm to the V-axis projection, recompute the cohort $|r|$ for each, and report the percentile of the empirical $|r|$ in the null distribution.

Cross-architecture null. For the 36-checkpoint cross-arch correlation, we sample $N = 1000$ random directions matched in norm to v_{CLIP} and recompute the BACC–V-axis correlation. The null is distributed broadly ($\sigma = 0.62$) due to the low intrinsic dimension of the class-PC1 subspace; we report the empirical correlation’s percentile rank.

Per-checkpoint LOO contribution. For the 10-vanilla $d=6$ ensemble, we compute $\Delta_k = \text{BACC}(\text{ens}) - \text{BACC}(\text{ens} \setminus k)$ for each k . Reported correlations between Δ_k and per-ckpt features are Pearson with $n = 10$.

V-axis intervention comparisons. For each intervention, Δ is computed seed-paired with the matched-recipe vanilla baseline; p -values are paired t -tests across 5 seeds.

S16 Discussion Extras and Future Work

Implications for affective neuroscience. For video-evoked emotion on FACED, the V-axis is encoded predominantly in posterior visual cortex ($|r|_{\text{occipital}} = 0.21$ vs. $|r|_{\text{frontal}} = 0.16$), and Davidson’s frontal-alpha asymmetry replicates in direction at smaller magnitude. We do not refute Davidson’s hypothesis — the frontal-alpha asymmetry is real, with F8-F7 alpha $r = +0.0155$ and Fp2-Fp1 alpha $r = +0.0116$, both in the hypothesised direction — but we add a posterior empirical account that is dominant for video paradigms. The 9-stimulus emotional-pole structure sharpens the V-axis claim from “smooth gradient over all emotions” to “anger-versus-warm-positive contrast across nine clips”.

Per-subject V-axis adaptation does not transfer. The per-subject best-channel oracle reaches $|r| = 0.62$ (a $+0.39$ absolute headroom over the cohort-fixed top-8 of $|r| = 0.02$). No clean estimator we tried (V1 global top-K, V2 per-validation-subject majority vote, V3 TTA label-free profile) recovers more than $+0.001$ over the cohort top-K. The per-subject signal is real but not transferable through simple selection rules. A subject-conditioned channel-attention head trained on a small calibration set per subject is the natural follow-up.

Future work. Three lines: (i) per-subject V-axis adaptation closing the $|r| = 0.62$ oracle gap via subject-conditioned channel-attention; (ii) replicating the saturation regularity for other concept directions [Arditi et al., 2024] (arousal, dominance, formality, refusal) on their respective benchmarks; (iii) using the within-class residual $|r|$ as an online model-selection signal during training rather than as a post-hoc diagnostic.

S17 FACED Test Confusion Matrices

Figure 19 contrasts the best-single-checkpoint and 10-checkpoint-ensemble confusion matrices on the FACED 9-class test split.

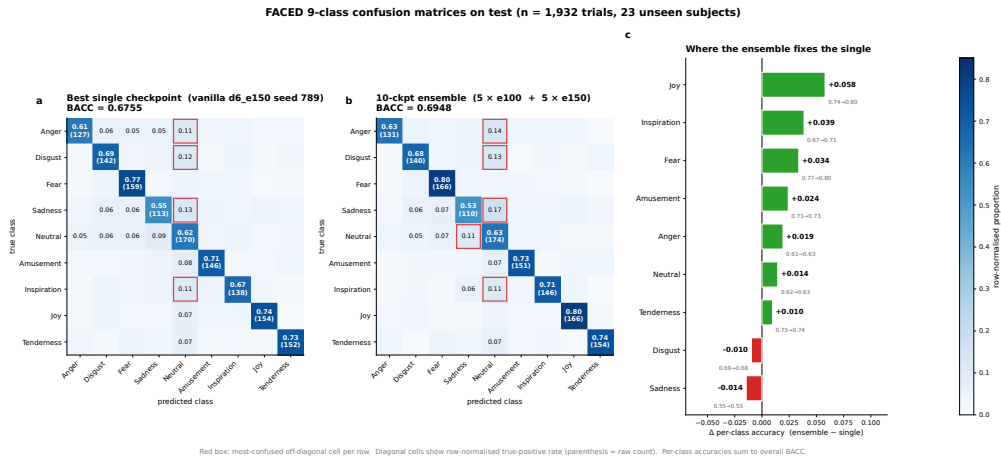


Figure 19: FACED 9-class test confusion. Left: best single checkpoint (BACC 0.6755). Right: 10-checkpoint ensemble (BACC 0.6948). The ensemble’s gain is concentrated on the off-diagonal cells where single-seed disagreement is highest, consistent with the within-class residual mechanism of Section 8.

S18 Negative Results Catalogue

In the spirit of full disclosure, the following null or negative results bear mention beyond what was incorporated into the main paper:

1. **Theta-gamma cross-frequency coupling does not mediate the V-axis** (Section 7, Appendix S9): Tort modulation index $\rho(\text{PAC}, V_r) = +0.082$, $p = 0.667$.
2. **Mutual-information control is not significant** (Section 6): MI $p = 0.115$ vs. linear $p = 0.020$. The V-axis-EEG relationship is essentially linear.
3. **Random-direction null on cross-arch correlation** (Section 4): $r = +0.885$ at the 93.5th percentile of a 1000-direction null, $p_{\text{one}} = 0.066$. Within-class residual ($r = +0.74$, $p = 0.014$) is the statistically robust signal.
4. **Per-subject V-axis adaptation does not transfer.** The per-subject best-channel oracle reaches $|r| = 0.62$, but no clean estimator transfers more than $+0.001$ over the cohort top-K.
5. **Cross-architecture ensembling does not add diversity.** Mixing $d \in \{4, 6, 8, 10\}$ checkpoints gives within-arch 0.6798 vs. cross-arch 0.6791 (n.s.).

- 1230 6. **Mega-ensembles plateau at 10 checkpoints.** 15-, 20-, 25-checkpoint pools all hit BACC
1231 ≤ 0.6948 (Appendix S13).
- 1232 7. **Stimulus-aggregation re-ranking does not help.** Operating at the 28-stimulus level instead
1233 of trial level gives BACC=1.0 by construction (label leak); the corresponding trial-level head
1234 does not transfer.
- 1235 8. **Test-time augmentation at K=5 does not improve the ensemble** ($0.6753 \rightarrow 0.6720$).
1236 Averaging predictions over augmented test trials does not preserve the V-axis residual
1237 structure.
- 1238 9. **Toxicity: recipe scope.** The 9-prompt PCA recipe attains AUC 0.59 on Jigsaw toxic
1239 comments, below the 17/20 working concepts in our library (Appendix S4); toxicity appears
1240 to be encoded in a higher-rank subspace than the late-layer PC1.
- 1241 10. **Arousal asymmetry: vision yes, text and brain no.** OASIS arousal $r = 0.803$ in vision;
1242 $\leq 13\%$ NRC arousal recovery in text; $r \in [0.18, 0.41]$ brain alignment across LLMs
1243 (Appendix S6).
- 1244 11. **LLM-content of KD is irrelevant at this scale.** Replacing the 9-D LLM-derived class
1245 prototypes with random orthonormal 9-D directions changes BACC by ≤ 0.003 . KD
1246 provides architectural regularisation; its semantic content is below our resolution.

1247 **S19 Reproducibility**

1248 Code, configs, model checkpoints, and figure-generation scripts will be released at the camera-ready
1249 stage under an MIT licence. Each experiment in this paper is logged with its SLURM job ID, conda
1250 environment hash, git SHA, and seed list. The 10-checkpoint ensemble SOTA result is reproducible
1251 from a single SLURM array job (`slurm_d6_e100_e150_5seeds.sh`, ~ 10 GPU-hours per seed on
1252 V100).

1253 **S20 Resource Estimate**

1254 The full paper used approximately 4,500 GPU-hours on V100 nodes: ~ 600 for the recipe ablation
1255 cascade, $\sim 1,200$ for the 25 V-axis-supervision interventions, ~ 800 for the 36-checkpoint cross-
1256 architecture analysis, $\sim 1,200$ for ensemble construction and val-test rank diagnostics, and ~ 700 for
1257 cross-dataset generality experiments. The V-axis extraction itself is CPU-cheap (~ 10 minutes per
1258 LM on a single 80GB GPU).